Quantum Information Processing (2026) 25:207
https://doi.org/10.1007/s11128-026-05230-0

Quantum codes construction from cyclic codes over matrix rings

Soham Ravikant Joshi[1] · Shikha Patel[2] · Om Prakash[1]

Received: 31 December 2025 / Accepted: 22 May 2026 / Published online: 9 June 2026
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2026

Abstract
Let $\mathbb{F}_p$ be the finite field with p elements where p is a prime, and $\mathcal{R}$ be a 3×3 matrices over $\mathbb{F}_p[u]/\langle u^k \rangle$, i.e., $\mathcal{R} = M_3(\mathbb{F}_p + u\mathbb{F}_p + \cdots + u^{k-1}\mathbb{F}_p)$ with $u^k = 0$. First, we study the structure of cyclic codes over $\mathcal{R}$ as the direct sum of submodules of $\mathcal{R}$, along with the Euclidean and Hermitian dual for the ring $\mathcal{S} = M_3(\mathbb{F}_3 + u\mathbb{F}_3 + \cdots + u^{k-1}\mathbb{F}_3)$. Later, to show that the structure is suitable for the construction of good linear codes, we provide many AMDS (almost maximum-distance-separable) and best-known linear codes over $\mathbb{F}_{125}$, $\mathbb{F}_{27}$, and $\mathbb{F}_8$. Further, we study dual-containing codes over $\mathcal{S}$ using a new non-degenerate inner product and provide a new approach to constructing quantum codes over matrix rings via the CSS (Calderbank-Shor-Steane) construction. Finally, we construct several quantum codes with small quantum Singleton defects using a suitable Gray map.

Keywords Modules · Matrix rings · Cyclic codes · Gray map · CSS construction · Quantum codes

Mathematics Subject Classification 12E20 · 94B05 · 94B15 · 81P45 · 81P70

1 Introduction

Cyclic codes are a special class of linear codes used for error detection and correction. They rely on polynomial representations over finite fields and rings, enabling

✉ Om Prakash
om@iitp.ac.in

Soham Ravikant Joshi
soham_2421ma14@iitp.ac.in

Shikha Patel
shikha_1821ma05@iitp.ac.in

[1] Department of Mathematics, Indian Institute of Technology Patna, Bihta, Patna 801 106, India

[2] Department of Mathematics, Indian Institute of Information Technology Bhopal, Bhopal 462003, India

efficient shift-register implementation and effective burst-error detection. Because of these advantages, cyclic codes have been widely studied in coding theory. Since 1994, codes over finite rings, including cyclic codes, have attracted considerable attention from researchers [4, 13, 14] and led to the construction of improved linear codes. Further, cyclic codes have been extensively investigated over both finite commutative and noncommutative rings. For instance, cyclic codes over finite commutative rings have been studied in [15, 27] to obtain distinct classes of codes. Meantime, several works have also explored these codes over matrix rings, see [17, 24, 26]. In fact, like skew polynomial rings, matrix rings also generate new and good linear codes. Towards this, Alahmadi et al. [2] obtained various self-dual $\mathbb{F}_4$-codes and cyclic codes over $M_2(\mathbb{F}_2)$ under Euclidean inner product. Later, the algebraic properties of cyclic codes over a more extended ring $M_2(\mathbb{F}_p)$ where $p \equiv 2$ or 3 (mod 5), were studied by Falcunit & Sison [9] in 2014. Additionally, in 2018, cyclic codes over the matrix ring $M_2(\mathbb{F}_2[u]/\langle u^2\rangle)$ were studied by Luo & Udaya [21], in which an isometry between $M_2(\mathbb{F}_2 + u\mathbb{F}_2)$ and $\mathbb{F}_4[u, v]/\langle u^2, v^2, uv - vu\rangle$ has been used to obtain various quaternary optimal codes. In 2024, Islam et al. [17] studied the cyclic codes over $M_3(\mathbb{F}_p)$, where they found many good cyclic codes over $\mathbb{F}_8$. In 2022, Patel et al. [26] studied cyclic codes over $M_4(\mathbb{F}_2 + u\mathbb{F}_2)$ and recently in 2025, Joshi et al. [19] extended this work to the construction of new codes over $\mathbb{F}_{16}$ from cyclic codes over $M_4(\mathbb{F}_2 + u\mathbb{F}_2 + u^2\mathbb{F}_2)$, whereas Patel et al. [25] extended the work of [17] to the construction of codes over $\mathbb{F}_{p^3}$ from cyclic codes over $M_3(\mathbb{F}_p + u\mathbb{F}_p + u^2\mathbb{F}_p)$. Motivated by these works, we extend the work of [17] to the more general structure $M_3(\mathbb{F}_p + u\mathbb{F}_p + \cdots + u^{k-1}\mathbb{F}_p)$ to see whether the structure is suitable for the construction of linear codes over $\mathbb{F}_{125}$ and $\mathbb{F}_{27}$, along with $\mathbb{F}_8$ for more generalized length $3kn$. These studies not only deepen the understanding of cyclic codes over the matrix rings but also indicate potential applications beyond classical coding theory, including quantum error correction, where dual-containing cyclic codes play an important role.

In classical information theory, error correction occurs in bits, whereas in quantum information theory, error correction occurs in quantum states (qubits). Both are concerned with the fundamental problem of reliable transmission of information over a noisy channel. However, quantum error-correcting codes (QECCs) protect quantum information from errors caused by decoherence and other types of quantum noise. Shor [31] in 1995, and Steane [32] in 1996 independently constructed quantum error-correcting codes from classical ones. In 1998, Calderbank et al. [6] proposed a systematic method to obtain quantum codes from classical ones, popularly known as the CSS (Calderbank-Shor-Steane) construction. In recent years, the construction of quantum codes via the CSS method has continued to benefit from advances in dual-containing classical codes. In particular, researchers have used cyclic, quasi-cyclic, skew-cyclic, and constacyclic codes and their dual-containing properties to construct quantum codes with improved or competitive parameters via the CSS construction.

In 2009, Qian et al. [30] provided a construction for quantum error-correcting codes from cyclic codes over the finite ring $R = \mathbb{F}_2 + u\mathbb{F}_2$ and obtained infinite families of quantum error-correcting codes. In 2011, Kai and Zhu [20] studied quantum codes from cyclic codes over $R = \mathbb{F}_4 + u\mathbb{F}_4$, where $u^2 = 0$. Dertli et al. [8] in 2015 studied the quantum codes obtained from cyclic codes over $A_2 = \mathbb{F}_2 + u\mathbb{F}_2 + v\mathbb{F}_2 + uv\mathbb{F}_2$, $u^2 = u$, $v^2 = v$, $uv = vu$. Later, in 2020, Islam et al. [18] studied the quantum codes from

constacyclic and additive constacyclic codes, and in 2021, Alahmadi et al. [1] studied the constacyclic codes over a non-chain ring $R_m = \mathbb{F}_p + u\mathbb{F}_p + \cdots + u^{m-1}\mathbb{F}_p$. In 2022, Islam et al. [16] studied a family of constacyclic codes over a class of non-chain rings $\mathcal{A}_{q,r}$ and and new quantum codes, whereas in 2023, Zau et al. [28] derived new quantum codes from images of cyclic codes. Recently, Prakash et al. [29] in 2023 studied quantum and LCD codes from skew constacyclic codes over a finite non-chain ring, whereas in 2025, Prakash et al. [34] studied quantum codes from skew cyclic codes over mixed alphabets. Following a similar framework, our two main objectives for this work are as follows.

(1) In [17], authors have studied the cyclic codes over $M_3(\mathbb{F}_p)$ and found many good linear codes over $\mathbb{F}_8$ with length $3n$ such that $p \nmid n$. In this paper, we extend this work to the study of cyclic codes over $\mathcal{R} = M_3(\mathbb{F}_p + u\mathbb{F}_p + \cdots + u^{k-1}\mathbb{F}_p)$ to get linear codes over more generalized length, which is $3kn$ where $p \nmid n$ and also study whether this structure is capable of producing good linear codes over $\mathbb{F}_{125}$ for $p = 5$, $\mathbb{F}_{27}$ for $p = 3$, along with $\mathbb{F}_8$ for $p = 2$.
(2) We propose a novel approach for constructing quantum codes using the CSS construction, leveraging cyclic codes over non-chain, noncommutative matrix rings. To the best of our knowledge, this is the first study in the literature that explores quantum code construction from such matrix rings. Specifically, we investigate dual-containing codes over the ring $\mathcal{S} = M_3(\mathbb{F}_3 + u\mathbb{F}_3 + \cdots + u^{k-1}\mathbb{F}_3)$, and utilize a suitable Gray map to obtain dual-containing linear codes over $\mathbb{F}_{27}$. These linear codes are then employed in the CSS framework to construct quantum codes with desirable parameters.

Moreover, an advantage of having length $3kn$ over $3n$ is that it is not possible to get cyclic codes of some lengths using the ring $M_3(\mathbb{F}_p)$. For example, $p = 3$, cyclic codes of length 18 over $\mathbb{F}_{27}$ cannot be found by [17], as $3n = 18$ implies $n = 6$, which is not possible. However, we can obtain cyclic codes of length 18 by setting $3kn$ where $n = 2$ and $k = 3$. Concrete examples of these cyclic codes of length 18 over $\mathbb{F}_{27}$ are given in Sect. 5.2.

The rest of the paper is organized as follows: Sect. 2 provides the construction of $\mathcal{R} = M_3(\mathbb{F}_p + u\mathbb{F}_p + \cdots + u^{k-1}\mathbb{F}_p)$ and defines the Gray map for the correspondence between $\mathbb{F}_{p^3}$ and $\mathcal{R}$. Section 3 studies the cyclic code over $\mathcal{R}$ as the direct sum of submodules of $\mathcal{R}$, while Sect. 4 discusses the Euclidean and Hermitian duals of the ring $\mathcal{S} = M_3(\mathbb{F}_3 + u\mathbb{F}_3 + \cdots + \cdots + u^{k-1}\mathbb{F}_3)$. Section 5 includes some examples of non-trivial linear codes over $\mathbb{F}_{125}$, $\mathbb{F}_{27}$, and $\mathbb{F}_8$, whereas Sect. 6 studies the quantum codes via the CSS construction, including the CSS code framework and defines a new associated Gray map. Section 7 provides many new quantum codes with better parameters, while Sect. 8 concludes the paper.

2 Construction of $\mathcal{R} = M_3(\mathbb{F}_p + u\mathbb{F}_p + \cdots + u^{k-1}\mathbb{F}_p)$

For a prime $p \leq 100$, let $\mathbb{F}_p$ be the finite field of p elements, and $f(x) \in \mathbb{F}_p[x]$, an irreducible polynomial of degree 3. For instance, $f(x) = x^3 - x - 1$ is irreducible in $\mathbb{F}_p[x]$ for $p \in \{2, 3, 13, 29, 31, 41, 47, 71, 73\}$ as it has no roots in $\mathbb{F}_p$. Similarly,

$g(x) = x^3 + x + 1$ is irreducible in $\mathbb{F}_p[x]$ for $p \in \{2, 5, 7, 19, 41, 59, 71, 97\}$ and $h(x) = x^3 + 3x + 2$ is irreducible in $\mathbb{F}_p[x]$ for $p \in \{5, 7, 11, 31, 53, 73, 83, 97\}$. We can choose any of $f(x)$, $g(x)$, or $h(x)$ depending on appropriate primes to construct $\mathbb{F}_{p^3} = \mathbb{F}_p[\omega]$ which is a Galois field where $f(\omega) = 0$ or $g(\omega) = 0$ or $h(\omega) = 0$. The Galois group $G = Gal(\mathbb{F}_{p^3}/\mathbb{F}_p)$ is cyclic generated by Frobenius automorphism $\sigma : \mathbb{F}_{p^3} \longrightarrow \mathbb{F}_{p^3}$ defined by $\sigma(a) = a^p$. Using the construction given in the [17, 24] we have a cyclic algebra $\mathbb{A} = (\mathbb{F}_{p^3}/\mathbb{F}_p, \sigma, 1) = \mathbb{F}_{p^3} + e\mathbb{F}_{p^3} + e^2\mathbb{F}_{p^3}$ where $ae = e\sigma(a)$, and $e^3 = 1$ for all $a \in \mathbb{F}_{p^3}$. Here, $\mathbb{A}$ is a central simple algebra with $dim(\mathbb{A}) = 9$ and therefore, $\mathbb{A} \cong M_3(\mathbb{F}_p)$. It is easy to check that $\mathcal{R} = M_3(\mathbb{F}_p[u]/\langle u^k\rangle) \cong M_3(\mathbb{F}_p)[u]/\langle u^k\rangle$. Hence,

$$\begin{aligned}\mathcal{R} &= (\mathbb{F}_{p^3} + e\mathbb{F}_{p^3} + e^2\mathbb{F}_{p^3}) + u(\mathbb{F}_{p^3} + e\mathbb{F}_{p^3} + e^2\mathbb{F}_{p^3})\\ &+u^2(\mathbb{F}_{p^3} + e\mathbb{F}_{p^3} + e^2\mathbb{F}_{p^3}) + \cdots + u^{k-1}(\mathbb{F}_{p^3} + e\mathbb{F}_{p^3} + e^2\mathbb{F}_{p^3}),\\ \text{where } e^3 = 1, u^k = 0, u^i e^j &= e^j u^i \text{ for } i = \{1, \ldots, k-1\}, \text{ and } j = \{1, 2, 3\}. \quad (1)\end{aligned}$$

2.1 Gray map for linear codes

In the case of ring $\mathbb{F}_2 + u\mathbb{F}_2$, we recall from [4] that Lee weight is defined by $\omega_L(0) = 0, \omega_L(1) = \omega_L(1 + u) = 1, \omega_L(u) = 2$, and the Gray map from $(\mathbb{F}_2 + u\mathbb{F}_2)^n$ to $\mathbb{F}_2^{2n}$ sends $(a + ub)$ to $(b, a + b)$ for $a, b \in \mathbb{F}_2^n$. Now, we will extend this Gray map θ from $\mathcal{R}$ to $\mathbb{F}_{p^3}^{3k}$ by

$$\begin{aligned}&\theta\big[(d_1 + ed_2 + e^2d_3) + u(d_4 + ed_5 + e^2d_6) + u^2(d_7 + ed_8 + e^2d_9)\\ &\quad + \cdots + u^{k-1}(d_{3k-2} + ed_{3k-1} + e^2d_{3k})\big]\\ &\quad = (d_{3k}, d_{3k-1} + d_{3k}, d_{3k-2} + d_{3k}, \ldots, d_2 + d_{3k}, d_1 + d_2 + d_3 + \cdots + d_{3k}),\end{aligned}$$

where $d_i \in \mathbb{F}_{p^3}$ for $i = \{1, \ldots, 3k\}$ and θ is also distance preserving. Hence, for any element $g = (d_1 + ed_2 + e^2d_3) + u(d_4 + ed_5 + e^2d_6) + u^2(d_7 + ed_8 + e^2d_9) + \cdots + u^{k-1}(d_{3k-2} + ed_{3k-1} + e^2d_{3k})$ in $\mathcal{R}$, the extended Lee weight is defined as:

$$\begin{aligned}\omega_L(g) &= (\omega_H(d_{3k}) + \omega_H(d_{3k-1} + d_{3k}) + \omega_H(d_{3k-2} + d_{3k})\\ &+ \cdots + \omega_H(d_2 + d_{3k}) + \omega_H(d_1 + d_2 + d_3 + \cdots + d_{3k}))\end{aligned}$$

where $\omega_H(.)$ denotes the Hamming weight in $\mathbb{F}_{p^3}$. Later, we extend this Gray map $\phi : \mathcal{R} \to \mathbb{F}_{p^3}^{3k}$ componentwise to $\mathcal{R}^n$. If we take $c = (c_0, c_1, \ldots, c_{n-1})$ in $\mathcal{R}^n$, then the Lee weight in c is $\omega_L(c) = \omega_L(c_0) + \omega_L(c_1) + \cdots + \omega_L(c_{n-1})$. The Hamming weight of a codeword $\omega_H(c)$ is the number of non-zero components. Hamming distance for any two codewords c and c' is defined as $d_H(c, c') = \omega_H(c - c')$. The Hamming distance for the code $\mathcal{C}$ is $d_L(\mathcal{C}) = min\{d_L(c, c') \mid c \neq c' \forall\, c, c' \in \mathcal{C}\}$.

Theorem 1 *Let $\mathcal{C}$ be a cyclic code over $\mathcal{R}$ of length n, size M, and Lee distance d. Then $\theta(\mathcal{C})$ is a linear code over $\mathbb{F}_{p^3}$ of length $3kn$, size M, and Hamming distance d.*

3 Cyclic codes over $M_3(\mathbb{F}_p + u\mathbb{F}_p + u^2\mathbb{F}_p + \cdots + u^{k-1}\mathbb{F}_p)$

A linear code $\mathcal{C}$ of length n over $\mathcal{R}$ is an $\mathcal{R}$-submodule of $\mathcal{R}^n$, and is said to be cyclic if for each codeword $c = (c_0, c_1, \ldots, c_{n-1}) \in \mathcal{C}$, its cyclic shift $(c_{n-1}, c_0, \ldots, c_{n-2}) \in \mathcal{C}$. A cyclic code of length n over $\mathcal{R}$ is also an ideal of $\mathcal{R}_n = \mathcal{R}[x]/\langle x^n - 1\rangle$ where each codeword $(c_0, c_1, \ldots, c_{n-1})$ corresponds to a polynomial $c_0 + c_1 x + \cdots + c_{n-1}x^{n-1}$. We restrict ourselves to cyclic codes of length n which satisfy $\gcd(n, p) = 1$, i.e., $p \nmid n$.

Let $\mathcal{R}[x]$ be the ring of polynomials over $\mathcal{R}$. For any $a \in \mathcal{R}$, $\overrightarrow{a}$ be the polynomial reduction of a modulo u and e as defined in Equation (1). We consider a polynomial reduction mapping $\Delta : \mathcal{R}[x] \to \mathbb{F}_{p^3}[x]$ defined by

$$\sum_{i=0}^{n-1} a_i x^i \mapsto \sum_{i=0}^{n-1} \overrightarrow{a_i}\, x^i.$$

Then a polynomial $f(x)$ is called basic irreducible over $\mathcal{R}$ if $\Delta(f(x))$ is irreducible over $\mathbb{F}_{p^3}$. If n is such that $p \nmid n$ and $x^n - 1 = \prod_{r=1}^{m} f_r$, where f_r's are irreducible polynomials in indeterminate x over $\mathbb{F}_{p^3}$, then the polynomial $x^n - 1$ can be uniquely factorized into pairwise coprime polynomials over $\mathbb{F}_{p^3}$.

Theorem 2 *Let n be such that $p \nmid n$ and $x^n - 1 = \prod_{r=1}^{m} f_r$, where f_r's are basic irreducible polynomials over $\mathcal{R}$. Then*

$$\mathcal{R}_n = \mathcal{R}[x]/\langle x^n - 1\rangle = \bigoplus_{r=1}^{m} \mathcal{R}[x]/\langle f_r\rangle.$$

Proof It follows by applying the Chinese Remainder Theorem on the right $\mathcal{R}$ - module $\mathcal{R}[x]/\langle x^n - 1\rangle$ [24, 33]. □

Since $\langle f_r\rangle$ is not a two-sided ideal, we consider $\mathcal{R}[x]/\langle f_r\rangle$ as a right $\mathcal{R}$-module. Now, we find all the possible forms of the submodules of $\mathcal{R}[x]/\langle f_r\rangle$.

Theorem 3 *Let f be a basic irreducible polynomial over $\mathcal{R}$. Then following is the complete set of right $\mathcal{R}$-submodules of $\mathcal{R}_f = \mathcal{R}[x]/\langle f\rangle$.*

1. $\{0\}$, $\langle u^i e^j\rangle$ *for* $i = \{0, 1, 2, \ldots, k-1\}$, $j = \{0, 1, 2\}$.
2. $\langle u^i + e\eta_{2i} + e^2\eta_{3i}\rangle$ *and* $\langle u^i, e, e^2\rangle$ *for* $i = \{1, 2, \ldots, k-1\}$,

where η_{mn} are units in $\mathbb{F}_{p^3}[x]$.

Proof Let $g + \langle f\rangle$ be a non-zero element of the $\mathcal{R}$-submodule I of $\mathcal{R}_f$. Then $g \notin \langle f\rangle$ and there exist g_{ij} for $i = \{1, \ldots, k\}$ and $j = \{1, 2, 3\}$ in $\mathbb{F}_{p^3}$ such that

$$g = \sum_{i=1}^{k} u^{i-1} g_{1i} + e \sum_{i=1}^{k} u^{i-1} g_{2i} + e^2 \sum_{i=1}^{k} u^{i-1} g_{3i}.$$

Now, we consider the following two cases.

(i) If we consider $\gcd(g_{11}, f) = 1$, then there exists $g_{11}^{-1} \in \mathbb{F}_{p^3}[x]$ such that $g_{11}g_{11}^{-1} + \langle f\rangle = 1 + \langle f\rangle$. Thus, $I = \mathcal{R}_f$.
(ii) If we consider $\gcd(g_{11}, f) = f$, then

$$g + \langle f\rangle = \sum_{i=2}^{k} u^{i-1}g_{1i} + e\sum_{i=1}^{k} u^{i-1}g_{2i} + e^2\sum_{i=1}^{k} u^{i-1}g_{3i}.$$

From case (ii), if we consider $\gcd(g_{1l}, f) = \gcd(g_{2l}, f) = \gcd(g_{3l}, f) = f$ for $l = \{1, \ldots, i\}$, and $\gcd(g_{1(i+1)}, f) = 1$, we have $g_{1(i+1)}^{-1}$ such that $g_{1(i+1)}g_{1(i+1)}^{-1} = 1$, and we have $I = \langle u^i\rangle$ for $i = \{1, \ldots, k-1\}$. If we consider $\gcd(g_{1l}, f) = f$ for $l = \{1, \ldots, k\}$, $\gcd(g_{2j}, f) = \gcd(g_{3j}, f) = f$ for $j = \{1\ldots., i\}$, and $\gcd(g_{2(i+1)}, f) = 1$, then there exists $g_{2(i+1)}^{-1}$ such that $f_{2(i+1)}f_{2(i+1)}^{-1} = 1$ and we have $I = \langle eu^i\rangle$. Following the similar process, we can show that $I = \langle e^2u^i\rangle$ for $i = \{1, \ldots, k-1\}$. If we consider $\gcd(g_{1l}, f) = f$ for $l = \{1\ldots., k\}$, and $\gcd(g_{21}, f) = 1$, there exist g_{21}^{-1} such that $g_{21}g_{21}^{-1} = 1$ and we have $I = \langle e\rangle$. Similarly, we can show that $I = \langle e^2\rangle$.

Now, let $\gcd(g_{1j}, f) = f$ for $j \in \{2, \ldots, i, i+2, \ldots, k\}$, and $\gcd(g_{2j}, f) = \gcd(g_{3j}, f) = f$ for $j \in \{2, \ldots, k\}$. If we consider $\gcd(g_{l1}, f) = 1$ for $l \in \{2, 3\}$ with $\gcd(g_{1(i+1)}, f) = 1$, then there exists $g_{1(i+1)}^{-1}$ such that $g_{1(i+1)}g_{1(i+1)}^{-1} = 1$ which provides

(I) $\langle u^i + e\eta_{2i} + e^2\eta_{3i}\rangle = I$.
(II) $\langle u^i + e\eta_{2i} + e^2\eta_{3i}\rangle \lneq I$.

Here, $\eta_{2i} = g_{21}g_{1(i+1)}^{-1}$, and $\eta_{3i} = g_{31}g_{1(i+1)}^{-1}$. Now in case (II), consider the following subcases.

(i) If $u^i + e\eta'_{2i} + e^2\eta_{3i} \in I$ where $\eta'_{2i} \in \mathbb{F}_{16}[x]$ such that it does not belong to $\langle u^i + e\eta_{2i} + e^2\eta_{3i}\rangle$. This implies that $e(\eta_{2i} - \eta'_{2i}) \in I$. But $(\eta_{2i} - \eta'_{2i}) \notin I$. Hence, $e \in I$. Thus, $I = \langle u^i, e\rangle$.
(ii) If there exists $u^i + e\eta_{2i} + e^2\eta'_{3i} \in I$ where $\eta'_{3i} \in \mathbb{F}_{16}[x]$ such that it does not belong to $\langle u^i + e\eta_{2i} + e^2\eta_{3i}\rangle$, then $e^2(\eta_{3i} - \eta'_{3i}) \in I$. But $(\eta_{3i} - \eta'_{3i}) \notin I$. Hence, $e^2 \in I$, and $I = \langle u^i, e^2\rangle$.

Thus, from subcases (i) and (ii), $I = \langle u^i, e, e^2\rangle$. □

Consider $\hat{g_i} = \frac{x^n-1}{g_i}$ where g_i is a factor of $x^n - 1$. Now, we have the following result for $\mathcal{R}$-modules in $\frac{\mathcal{R}[x]}{\langle x^n-1\rangle}$.

Corollary 1 *Any right $\mathcal{R}$-module in $\frac{\mathcal{R}[x]}{\langle x^n-1\rangle}$ is sum of the following $\mathcal{R}$-submodules for $x^n - 1 = g_1g_2\cdots g_m$, where $g_j (1 \le j \le m)$ are irreducible pairwise coprime monic polynomials in $\mathbb{F}_{p^3}[x]$.*

1. $\langle u^{i-1}e^l\hat{g_j} + \langle x^n - 1\rangle\rangle$ *for* $i = \{1, \ldots, k\}$, *and* $l = \{0, 1, 2\}$.
2. $\langle (u^i + e\eta_{1i} + e^2\eta_{2i})\hat{g_j} + \langle x^n - 1\rangle\rangle$ *for* $i = \{1, \ldots, k\}$.
3. $(\langle u^i\hat{g}_j + \langle x^n - 1\rangle\rangle + \langle e\hat{g}_j + \langle x^n - 1\rangle\rangle + \langle e^2\hat{g}_j + \langle x^n - 1\rangle\rangle)$ *for* $i = \{1, \ldots, k\}$.

Proof This follows from Theorem 2 and Theorem 3. □

For the ring $\frac{\mathcal{R}[x]}{\langle x^n-1\rangle}$, we use $a_0+a_1x+\cdots+a_{n-1}x^{n-1}$ for the corresponding coset $a_0+a_1x+\cdots+a_{n-1}x^{n-1}+\langle x^n-1\rangle$.

Theorem 4 *Let $\mathcal{P}$ be a family of irreducible pairwise coprime monic polynomials P_i ($0\leq i\leq 4k-1$) in $\mathbb{F}_{p^3}[x]$ such that $\prod_{i=0}^{4k-1}P_i=x^n-1$. Then a cyclic code $\mathcal{C}$ over $\mathcal{R}$ can be written as*

$$\mathcal{C}=\bigoplus_{i=1}^{k}\langle u^{i-1}\hat{P}_i\rangle\bigoplus_{i=1}^{k}\langle u^{i-1}e\hat{P}_{k+i}\rangle\bigoplus_{i=1}^{k}\langle u^{i-1}e^2\hat{P}_{2k+i}\rangle\bigoplus_{i=1}^{k-1}\langle(u^i+e\eta_{1i}+e^2\eta_{2i})\hat{P}_{3k+i}\rangle.$$

Moreover, $|\mathcal{C}|=p^{3\gamma}$ where

$$\gamma=\left(\sum_{i=1}^{k}(3k-3(i-1))\deg(P_i)+(2k-2(i-1))\deg(P_{k+i})+(k-(i-1))\deg(P_{2k+i})\right)+\left(\sum_{i=1}^{k-1}(3k-i)\deg(P_{3k+i})\right).$$

Proof Let $\mathcal{G}$ be a family of irreducible pairwise coprime monic polynomials g_i in $\mathbb{F}_{p^3}[x]$ such that $\prod_{i=1}^{m}g_i=x^n-1$. Then from the Corollary 1, any right $\mathcal{R}$-module in $\frac{\mathcal{R}[x]}{\langle x^n-1\rangle}$ is sum of the following $\mathcal{R}$-submodules.

1. $\langle u^{i-1}e^l\hat{g_j}+\langle x^n-1\rangle\rangle$ for $i=\{1,\ldots,k\}$, and $l=\{0,1,2\}$.
2. $\langle(u^i+e\eta_{1i}+e^2\eta_{2i})\hat{g_j}+\langle x^n-1\rangle\rangle$ for $i=\{1,\ldots,k\}$.

Without loss of generality, we can assume that

$$\mathcal{C}=\bigoplus_{i=1}^{k}\langle u^{i-1}\hat{f}_{m_1+\cdots+m_i+1}\rangle\oplus\cdots\oplus\langle u^{i-1}\hat{f}_{m_1+\cdots+m_i+m_{i+1}}\rangle$$

$$\bigoplus_{i=1}^{k}\langle u^{i-1}e\hat{f}_{m_1+\cdots+m_{k+i}+1}\rangle\oplus\cdots\oplus\langle u^{i-1}e\hat{f}_{m_1+\cdots+m_{k+(i+1)}}\rangle$$

$$\bigoplus_{i=1}^{k}\langle u^{i-1}e^2\hat{f}_{m_1+\cdots+m_{2k+i}+1}\rangle\oplus\cdots\oplus\langle u^{i-1}e^2\hat{f}_{m_1+\cdots+m_{2k+i+1}}\rangle$$

$$\bigoplus_{i=1}^{k-2}\langle(u^i+e\eta_{1i}+e^2\eta_{2i})\hat{f}_{m_1+\cdots+m_{3k+i}+1}\rangle\bigoplus\cdots\bigoplus\langle(u^i+e\eta_{1i}+e^2\eta_{2i})\hat{f}_{m_1+\cdots+m_{3k+i+1}}\rangle$$

$$\bigoplus \langle (u^{k-1} + e\eta_{1(k-1)} + e^2\eta_{2(k-1)})\hat{f}_{m_1+\cdots+m_{4k-1}+1}\rangle$$
$$\bigoplus \cdots \bigoplus \langle (u^{k-1} + e\eta_{1(k-1)} + e^2\eta_{2(k-1)})\hat{f}_{m}\rangle$$

where integers $m_1, m_2, m_3, \ldots, m_{4k-1} \geq 0$ and $m_1 + m_2 + \cdots + m_{4k-1} + 1 \leq m$. Let $m_0 = 0$ and m_{4k} be the integer such that $m_1 + m_2 + \cdots + m_{4k} = m$. Now, we define

$$P_0 = f_{m_0+1} \cdots f_{m_0+m_1},\ P_1 = f_{m_0+m_1+1} \cdots f_{m_0+m_1+m_2},$$
$$P_2 = f_{m_0+m_1+m_2+1} \cdots f_{m_0+m_1+m_2+m_3},\ P_3 = f_{m_0+\cdots+m_3+1} \cdots f_{m_0+\cdots+m_4},$$
$$\ldots,\ P_{4k-1} = f_{m_0+\cdots+m_{4k-1}+1} \cdots f_m.$$

From the construction above, we can see that $P_0, P_1, \ldots, P_{4k-1}$ are pairwise coprime and $\prod_{i=0}^{4k-1} P_i = x^n - 1$. Hence,

$$\mathcal{C} = \bigoplus_{i=1}^{k} \langle u^{i-1}\hat{P}_i\rangle \bigoplus_{i=1}^{k} \langle u^{i-1}e\hat{P}_{k+i}\rangle \bigoplus_{i=1}^{k} \langle u^{i-1}e^2\hat{P}_{2k+i}\rangle \bigoplus_{i=1}^{k-1} \langle (u^i + e\eta_{1i} + e^2\eta_{2i})\hat{P}_{3k+i}\rangle.$$

Furthermore,

$$|\langle u^{i-1}\hat{P}_i\rangle| = p^{3(3k-3(i-1))\deg(P_i)}, |\langle u^{i-1}e\hat{P}_{k+i}\rangle| = p^{3(2k-2(i-1))\deg(P_{k+i})},$$
$$|\langle u^{i-1}e^2\hat{P}_{2k+i}\rangle| = p^{3(k-(i-1))\deg(P_{2k+i})},\ \text{for } i = \{1, \ldots, k\}.$$
$$|\langle (u^i + e\eta_{1i} + e^2\eta_{2i})\hat{P}_{3k+i}\rangle| = p^{3(3k-i)\deg(P_{3k+i})},\ \text{for } i = \{1, \ldots, k-1\}.$$

Thus, $|\mathcal{C}| = p^{3\gamma}$ where

$$\gamma = \left(\sum_{i=1}^{k} (3k - 3(i-1))\deg(P_i) + (2k - 2(i-1))\deg(P_{k+i}) + (k - (i-1))\deg(P_{2k+i})\right)$$
$$+ \left(\sum_{i=1}^{k-1} (3k - i)\deg(P_{3k+i})\right).$$

□

Now, let $\langle x^n - 1\rangle_{\mathbb{F}_{p^3}[x]}$ be an ideal of $\mathbb{F}_{p^3}[x]$ generated by $x^n - 1$. We consider $\mathcal{R}' = \frac{\mathbb{F}_{p^3}[x]}{\langle x^n-1\rangle_{\mathbb{F}_{p^3}[x]}}$, and we have the following theorem.

Theorem 5 *For a cyclic code $\mathcal{C}$ of a length n over $\mathcal{R}$, there exist polynomials $C_1, C_2, C_3, \ldots, C_{4k-1}$ in $\mathbb{F}_{p^3}[x]$ which are factors of $x^n - 1$ such that*

$$\mathcal{C} = \bigoplus_{i=1}^{k} u^{i-1}\langle C_i\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{k} u^{i-1}e\langle C_{k+i}\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{k} u^{i-1}e^2\langle C_{2k+i}\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{k-1} (u^i + e\eta_{1i} + e^2\eta_{2i})\langle C_{3k+i}\rangle_{\mathcal{R}'}$$

where η_{mn} are units in $\mathbb{F}_{p^3}[x]$ and $\langle - \rangle_{\mathcal{R}'}$ is an ideal of $\mathcal{R}'$ generated by $-$. Also,

$$|\mathcal{C}| = p^{3((4k-1)n-(\deg C_1+\deg C_2+\cdots \deg C_{4k-1}))}.$$

Proof From Theorem 4, we have

$$\mathcal{C} = \bigoplus_{i=1}^{k} \langle u^{i-1}\hat{P}_i\rangle \bigoplus_{i=1}^{k} \langle u^{i-1}e\hat{P}_{k+i}\rangle \bigoplus_{i=1}^{k} \langle u^{i-1}e^2\hat{P}_{2k+i}\rangle \bigoplus_{i=1}^{k-1} \langle (u^i + e\eta_{1i} + e^2\eta_{2i})\hat{P}_{3k+i}\rangle.$$

We can easily show that

$$\langle \hat{P}_i\rangle = \bigoplus_{j=1}^{k} u^{j-1}\langle \hat{P}_i\rangle_{\mathcal{R}'} \bigoplus_{j=1}^{k} u^{j-1}e\langle \hat{P}_i\rangle_{\mathcal{R}'} \bigoplus_{j=1}^{k} u^{j-1}e^2\langle \hat{P}_i\rangle_{\mathcal{R}'}.$$

For $0 \le i \le 4k-1$, we have

$$\mathcal{C} = \bigoplus_{i=1}^{k} u^{i-1}\left(\bigoplus_{j=1}^{(k-1)-(i-2)} u^{j-1}\langle \hat{P}_i\rangle_{\mathcal{R}'} \oplus u^{j-1}e\langle \hat{P}_i\rangle_{\mathcal{R}'} \oplus u^{j-1}e^2\langle \hat{P}_i\rangle_{\mathcal{R}'}\right)$$

$$\bigoplus_{i=1}^{k} u^{i-1}e\left(\bigoplus_{j=1}^{(k-1)-(i-2)} u^{j-1}\langle \hat{P}_{k+i}\rangle_{\mathcal{R}'} \oplus u^{j-1}e\langle \hat{P}_{k+i}\rangle_{\mathcal{R}'}\right)$$

$$\bigoplus_{i=1}^{k} u^{i-1}e^2\left(\bigoplus_{j=1}^{(k-1)-(i-2)} u^{j-1}\langle \hat{P}_{2k+i}\rangle_{\mathcal{R}'}\right)$$

$$\bigoplus_{i=1}^{k-1}(u^i + e\eta_{1i} + e^2\eta_{2i})\left(\bigoplus_{j=1}^{(k-1)-(i-2)} u^{j-1}\langle \hat{P}_{3k+i}\rangle_{\mathcal{R}'} \oplus u^{j-1}e\langle \hat{P}_{3k+i}\rangle_{\mathcal{R}'} \oplus u^{j-1}e^2\langle \hat{P}_{3k+i}\rangle_{\mathcal{R}'}\right).$$

$$= \langle \hat{P}_1\rangle_{\mathcal{R}'} \bigoplus_{j=1}^{k-1} u^j \left(\bigoplus_{i=1}^{j+1}\langle \hat{P}_i\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{j}\langle \hat{P}_{3k+i}\rangle_{\mathcal{R}'}\right) \bigoplus_{j=1}^{2} e^j \left(\bigoplus_{i=1}^{j}\langle \hat{P}_{ik+1}\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{k-1}\langle \hat{P}_{3k+i}\rangle_{\mathcal{R}'} \bigoplus\langle \hat{P}_1\rangle_{\mathcal{R}'}\right)$$

$$\bigoplus_{j=1}^{k-1} u^j e\left(\bigoplus_{i=1}^{j+1}\langle \hat{P}_i\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{j+1}\langle \hat{P}_{k+i}\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{k-1}\langle \hat{P}_{3k+i}\rangle_{\mathcal{R}'}\right)$$

$$\bigoplus_{j=1}^{k-1} u^j e^2\left(\bigoplus_{i=1}^{j+1}\langle \hat{P}_i\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{j+1}\langle \hat{P}_{k+i}\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{j+1}\langle \hat{P}_{2k+i}\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{k-1}\langle \hat{P}_{3k+i}\rangle_{\mathcal{R}'}\right)$$

$$\bigoplus_{j=1}^{k-1}(u^i + e\eta_{1i} +^2 \eta_{2i}) \left(\bigoplus_{i=1}^{j}\langle\hat{P}_{3k+i}\rangle_{\mathcal{R}'}\right).$$

We consider

$$C_1 = \hat{P}_1,\ C_{j+1} = \left(\bigoplus_{i=1}^{j+1}\hat{P}_i \bigoplus_{i=1}^{j}\hat{P}_{3k+i}\right) \text{for } j = \{1, k-1\}.\ C_{k+1} = \left(\hat{P}_{k+1}\bigoplus_{i=1}^{k-1}\hat{P}_{3k+i} \bigoplus \hat{P}_1\right),$$

$$C_{k+1+j} = \left(\bigoplus_{i=1}^{j+1}\hat{P}_i \bigoplus_{i=1}^{j+1}\hat{P}_{k+i}\bigoplus_{i=1}^{k-1}\hat{P}_{3k+i}\right) \text{for } j = \{1, k-1\}.$$

$$C_{2k+1} = \left(\bigoplus_{i=1}^{j}\hat{P}_{ik+1}\bigoplus_{i=1}^{k-1}\hat{P}_{3k+i} \bigoplus \hat{P}_1\right) \text{for } j = 2.$$

$$C_{2k+1+j} = \left(\bigoplus_{i=1}^{j+1}\hat{P}_i \bigoplus_{i=1}^{j+1}\hat{P}_{k+i}\bigoplus_{i=1}^{j+1}\hat{P}_{2k+i}\bigoplus_{i=1}^{k-1}\hat{P}_{3k+i}\right) \text{for } j = \{1, k-1\}.$$

$$C_{3k+j} = \left(\bigoplus_{i=1}^{j}\hat{P}_{3k+i}\right) \text{for } j = \{1, k-1\}.$$

Now, we have to prove

$$\langle C_1\rangle_{\mathcal{R}'} = \langle\hat{P}_1\rangle_{\mathcal{R}'},\ \langle C_{j+1}\rangle_{\mathcal{R}'} = \left(\bigoplus_{i=1}^{j+1}\langle\hat{P}_i\rangle_{\mathcal{R}'}\bigoplus_{i=1}^{j}\langle\hat{P}_{3k+i}\rangle_{\mathcal{R}'}\right) \text{for } j = \{1, k-1\}.$$

$$\langle C_{k+1}\rangle_{\mathcal{R}'} = \left(\langle\hat{P}_{k+1}\rangle_{\mathcal{R}'}\bigoplus_{i=1}^{k-1}\langle\hat{P}_{3k+i}\rangle_{\mathcal{R}'}\bigoplus\langle\hat{P}_1\rangle_{\mathcal{R}'}\right),$$

$$\langle C_{k+1+j}\rangle_{\mathcal{R}'} = \left(\bigoplus_{i=1}^{j+1}\langle\hat{P}_i\rangle_{\mathcal{R}'}\bigoplus_{i=1}^{j+1}\langle\hat{P}_{k+i}\rangle_{\mathcal{R}'}\bigoplus_{i=1}^{k-1}\langle\hat{P}_{3k+i}\rangle_{\mathcal{R}'}\right) \text{for } j = \{1, k-1\}.$$

$$\langle C_{2k+1}\rangle_{\mathcal{R}'} = \left(\bigoplus_{i=1}^{j}\langle\hat{P}_{ik+1}\rangle_{\mathcal{R}'}\bigoplus_{i=1}^{k-1}\langle\hat{P}_{3k+i}\rangle_{\mathcal{R}'}\bigoplus\langle\hat{P}_1\rangle_{\mathcal{R}'}\right) \text{for } j = 2.$$

$$\langle C_{2k+1+j}\rangle_{\mathcal{R}'} = \left(\bigoplus_{i=1}^{j+1}\langle\hat{P}_i\rangle_{\mathcal{R}'}\bigoplus_{i=1}^{j+1}\langle\hat{P}_{k+i}\rangle_{\mathcal{R}'}\bigoplus_{i=1}^{j+1}\langle\hat{P}_{2k+i}\rangle_{\mathcal{R}'}\bigoplus_{i=1}^{k-1}\langle\hat{P}_{3k+i}\rangle_{\mathcal{R}'}\right) \text{for } j = \{1, k-1\}.$$

$$\langle C_{3k+j}\rangle_{\mathcal{R}'} = \left(\bigoplus_{i=1}^{j}\langle \hat{P}_{3k+i}\rangle_{\mathcal{R}'}\right) \text{ for } j = \{1, k-1\}.$$

Now, from the hypothesis, we have $x^n - 1 | \hat{P}_i \hat{P}_j$, and $\hat{P}_i \hat{P}_j = 0$ in $\frac{\mathcal{R}[x]}{\langle x^n-1\rangle}$ for any distinct $i, j \in \{1, \ldots, 4k-1\}$. As $\{P_j, \hat{P}_j\}$ are pairwise coprime, there exist $y_{0j}, y_{1j} \in \mathbb{F}_{p^3}[x]$ such that $y_{0j} P_j + y_{1j} \hat{P}_j = 1$ for $j = \{1, \ldots, 4k-1\}$. Further, if we consider $y_{0j} P_j + y_{1j} \hat{P}_j = 1$ for $j = \{2, \ldots, 4k-1\}$, there exist polynomials $u_1, u_2, \ldots, u_{4k-1}$ in $\mathbb{F}_{p^3}[x]$ such that

$$u_1 P_2 \cdots P_{4k-1} + u_2 \hat{P}_2 P_3 \cdots P_{4k-1} + \cdots + u_{4k-1} P_2 P_3 \cdots P_{4k-2} \hat{P}_{4k-1} = 1.$$

If we multiply the above equation by $\hat{P}_1$, we have $\hat{P}_1 = u_1 \hat{P}_1 P_2 \cdots P_{4k-1}$. From the above construction, we can conclude that $u_1 C_1 P_2 \cdots P_{4k-1} = \hat{P}_1$ and this implies that $\langle C_1\rangle_{\mathcal{R}'} = \langle \hat{P}_1\rangle$. Now, $C_2 = \hat{P}_1 + \hat{P}_2 + \hat{P}_{3k+1}$. Hence, $u_1 C_2 P_2 \cdots P_{4k-1} = \hat{P}_1$ which implies that $\hat{P}_1 \in \langle C_2\rangle_{\mathcal{R}'}$. If we continue this process, we get $\hat{P}_2 \in \langle C_2\rangle_{\mathcal{R}'}$, $\hat{P}_{3k+1} \in \langle C_2\rangle_{\mathcal{R}'}$. Hence, we can conclude that

$$\langle C_2\rangle_{\mathcal{R}'} = \langle \hat{P}_1\rangle_{\mathcal{R}'} + \langle \hat{P}_2\rangle_{\mathcal{R}'} + \langle \hat{P}_{3k+1}\rangle_{\mathcal{R}'}.$$

Similarly, we can show that

$$\langle C_1\rangle_{\mathcal{R}'} = \langle \hat{P}_1\rangle_{\mathcal{R}'}, \langle C_{j+1}\rangle_{\mathcal{R}'} = \left(\bigoplus_{i=1}^{j+1}\langle \hat{P}_i\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{j}\langle \hat{P}_{3k+i}\rangle_{\mathcal{R}'}\right) \text{ for } j = \{2, k-1\}.$$

$$\langle C_{k+1}\rangle_{\mathcal{R}'} = \left(\langle \hat{P}_{k+1}\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{k-1}\langle \hat{P}_{3k+i}\rangle_{\mathcal{R}'} \bigoplus \langle \hat{P}_1\rangle_{\mathcal{R}'}\right),$$

$$\langle C_{k+1+j}\rangle_{\mathcal{R}'} = \left(\bigoplus_{i=1}^{j+1}\langle \hat{P}_i\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{j+1}\langle \hat{P}_{k+i}\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{k-1}\langle \hat{P}_{3k+i}\rangle_{\mathcal{R}'}\right) \text{ for } j = \{1, k-1\}.$$

$$\langle C_{2k+1}\rangle_{\mathcal{R}'} = \left(\bigoplus_{i=1}^{j}\langle \hat{P}_{ik+1}\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{k-1}\langle \hat{P}_{3k+i}\rangle_{\mathcal{R}'} \bigoplus \langle \hat{P}_1\rangle_{\mathcal{R}'}\right) \text{ for } j = 2.$$

$$\langle C_{2k+1+j}\rangle_{\mathcal{R}'} = \left(\bigoplus_{i=1}^{j+1}\langle \hat{P}_i\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{j+1}\langle \hat{P}_{k+i}\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{j+1}\langle \hat{P}_{2k+i}\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{k-1}\langle \hat{P}_{3k+i}\rangle_{\mathcal{R}'}\right) \text{ for } j = \{1, k-1\}.$$

$$\langle C_{3k+j}\rangle_{\mathcal{R}'} = \left(\bigoplus_{i=1}^{j}\langle \hat{P}_{3k+i}\rangle_{\mathcal{R}'}\right) \text{ for } j = \{1, k-1\}.$$

Since $|\mathbb{F}_{p^3}| = p^3$, $|\langle C_j \rangle_{\mathcal{R}'}| = p^{3(n-\deg C_j)}$ and hence, $|\mathcal{C}| = p^{3((4k-1)n-(\sum_{j=1}^{4k-1} \deg C_j))}$. □

4 Dual of cyclic codes over $M_3(\mathbb{F}_3 + u\mathbb{F}_3 + u^2\mathbb{F}_3 + \cdots + u^{k-1}\mathbb{F}_3)$

In the study of linear codes over finite rings or fields with a chosen inner product, we define important structural properties such as self-orthogonality, dual-containing, and self-duality. A code C is said to be self-orthogonal if $C \subseteq C^\perp$, dual-containing if $C^\perp \subseteq C$, and self-dual if $C = C^\perp$, where $C^\perp$ denotes the dual of C. Studying these duals is particularly important because codes that are self-orthogonal or dual-containing play a fundamental role in constructing quantum error-correcting codes. In our case, we are studying the structure of such duals for the ring $\mathcal{S} = M_3(\mathbb{F}_3 + u\mathbb{F}_3 + u^2\mathbb{F}_3 + \cdots + u^{k-1}\mathbb{F}_3)$.

4.1 Structure of the ring $\mathcal{S}$

From the Equation (1), we have

$$\begin{aligned}\mathcal{R} = M_3(\mathbb{F}_p + u\mathbb{F}_p + \cdots + u^{k-1}\mathbb{F}_p) &\cong (\mathbb{F}_{p^3} + e\mathbb{F}_{p^3} + e^2\mathbb{F}_{p^3}) + u(\mathbb{F}_{p^3} + e\mathbb{F}_{p^3} + e^2\mathbb{F}_{p^3}) \\ &+ u^2(\mathbb{F}_{p^3} + e\mathbb{F}_{p^3} + e^2\mathbb{F}_{p^3}) + \cdots + u^{k-1}(\mathbb{F}_{p^3} + e\mathbb{F}_{p^3} + e^2\mathbb{F}_{p^3}), \\ &\text{where } e^3 = 1, u^k = 0, u^i e^j = e^j u^i \text{ for } i = \{1, \ldots, k-1\}, \text{ and } j = \{1, 2, 3\}.\end{aligned}$$

For $p = 3$, we have $(e-1)^3 = e^3 - 1 = 0$. Hence, if we put $v = e - 1$, we have

$$\begin{aligned}\mathcal{S} = M_3(\mathbb{F}_3 + u\mathbb{F}_3 + \cdots + u^{k-1}\mathbb{F}_3) &\cong (\mathbb{F}_{27} + v\mathbb{F}_{27} + v^2\mathbb{F}_{27}) + u(\mathbb{F}_{27} + v\mathbb{F}_{27} + v^2\mathbb{F}_{27}) \\ &+ u^2(\mathbb{F}_{27} + v\mathbb{F}_{27} + v^2\mathbb{F}_{27}) + \cdots + u^{k-1}(\mathbb{F}_{27} + v\mathbb{F}_{27} + v^2\mathbb{F}_{27}), \\ &\text{where } v^3 = 0, u^k = 0, u^i v^j = v^j u^i \text{ for } i = \{1, \ldots, k-1\}, \text{ and } j = \{1, 2, 3\}. \qquad (2)\end{aligned}$$

Remark 1 Note that studying dual codes is not always feasible in the general setting of cyclic codes over the ring $\mathcal{R}$. However, in the special case $\mathcal{S}$, the ring admits nontrivial nilpotent elements u and v. These nilpotent elements provide the necessary algebraic structure to investigate the duals of cyclic codes. In this direction, we first study the structure of $\mathcal{S}$ separately, followed by the construction of the dual over it.

Let $\mathcal{S}[x]$ be the ring of polynomials over $\mathcal{S}$. For any $a \in \mathcal{S}$, $\overrightarrow{a}$ is the polynomial reduction of a modulo u and v as defined in Equation (2). Then similar to the polynomial reduction mapping Δ defined in Sect. 3, a polynomial $f(x)$ is called basic irreducible over $\mathcal{S}$ if $\Delta(f(x))$ is irreducible over $\mathbb{F}_{27}$. If n is such that $3 \nmid n$ and $x^n - 1 = \prod_{r=1}^{m} f_r$ where f_r's are irreducible polynomials in indeterminate x over $\mathbb{F}_{27}$, then the polynomial $x^n - 1$ can be uniquely factorized into pairwise coprime polynomials over $\mathbb{F}_{27}$. Also, it follows from Theorem 2 that if n is such that $3 \nmid n$ and $x^n - 1 = \prod_{r=1}^{m} f_r$ where f_r's are basic irreducible polynomials over $\mathcal{S}$, then

$$\mathcal{S}_n = \mathcal{S}[x]/\langle x^n - 1\rangle = \bigoplus_{r=1}^{m} \mathcal{S}[x]/\langle f_r\rangle.$$

Since $\langle f_r\rangle$ is not a two-sided ideal, we consider $\mathcal{S}[x]/\langle f_r\rangle$ as a right $\mathcal{S}$-module. Now, we find all the possible forms of the submodules of $\mathcal{S}[x]/\langle f_r\rangle$.

Theorem 6 *Let f be a basic irreducible polynomial over $\mathcal{S}$. Then the following is the complete set of right $\mathcal{S}$-submodules of $\mathcal{S}_f = \mathcal{S}[x]/\langle f\rangle$.*

1. $\{0\}$, $\langle u^i v^j\rangle$ *for* $i = \{0, 1, 2, \ldots, k-1\}$, $j = \{0, 1, 2\}$.
2. $\langle u^i + v\eta_{2i} + v^2\eta_{3i}\rangle$ *and* $\langle u^i, v, v^2\rangle$ *for* $i = \{1, 2, \ldots, k-1\}$.
3. $\langle u + \sum_{i=3}^{k} u^{i-1}\beta_{1i} + v\sum_{i=1}^{k} u^{i-1}\beta_{2i} + v^2\sum_{i=1}^{k-1} u^{i-1}\beta_{3i}\rangle$.
4. $\langle uv + v\sum_{i=3}^{k} u^{i-1}\beta_{1i} + v^2\sum_{i=1}^{k-1} u^{i-1}\beta_{2i}\rangle$.
5. $\langle uv^2 + v^2\sum_{i=3}^{k-1} u^{i-1}\beta_{1i}\rangle$.

where β_{ij} and η_{mn} are units in $\mathbb{F}_{27}[x]$.

Proof Let $g + \langle f\rangle$ be a non-zero element of the $\mathcal{S}$-submodule I of $\mathcal{S}_f$. Then $g \notin \langle f\rangle$ and there exists g_{ij} for $i = \{1, \ldots, k\}$ and $j = \{1, 2, 3\}$ in $\mathbb{F}_{27}$ such that

$$g = \sum_{i=1}^{k} u^{i-1}g_{1i} + v\sum_{i=1}^{k} u^{i-1}g_{2i} + v^2\sum_{i=1}^{k} u^{i-1}g_{3i}.$$

Now, we consider the following two cases.

(i) If we take $\gcd(g_{11}, f) = 1$, then there exists $g_{11}^{-1} \in \mathbb{F}_{27}[x]$ such that $g_{11}g_{11}^{-1} + \langle f\rangle = 1 + \langle f\rangle$. Thus, $I = \mathcal{S}_f$.
(ii) If we take $\gcd(g_{11}, f) = f$, then

$$g + \langle f\rangle = \sum_{i=2}^{k} u^{i-1}g_{1i} + v\sum_{i=1}^{k} u^{i-1}g_{2i} + v^2\sum_{i=1}^{k} u^{i-1}g_{3i}.$$

Construction of right $\mathcal{S}$-submodules listed in 1., and 2. follows similar to the construction of the right $\mathcal{R}$-submodules 1, and 2 in the Theorem 3.

Now, from case (ii), if we consider $\gcd(g_{12}, f) = 1$, we have g_{12}^{-1} such that $u^{k-1}v^2 = gu^{k-2}v^2g_{12}^{-1} \in I$, and

$$\sum_{i=2}^{k} u^{i-1}g_{1i} + v\sum_{i=1}^{k} u^{i-1}g_{2i} + v^2\sum_{i=1}^{k-1} u^{i-1}g_{3i} = g - v^2u^{k-1}g_{3k}.$$

If we consider $\gcd(g_{1l}, f) = 1$ for $l = \{3, \ldots, k\}$, $\gcd(g_{2j}, f) = 1$ for $j = \{1, \ldots, k\}$, and $\gcd(g_{3m}, f) = 1$ for $m = \{1, \ldots, k-1\}$, we have the following cases:

1. $\langle u + \sum_{i=3}^{k} u^{i-1}\beta_{1i} + v\sum_{i=1}^{k} u^{i-1}\beta_{2i} + v^2\sum_{i=1}^{k-1} u^{i-1}\beta_{3i}\rangle = I$.
2. $\langle u + \sum_{i=3}^{k} u^{i-1}\beta_{1i} + v\sum_{i=1}^{k} u^{i-1}\beta_{2i} + v^2\sum_{i=1}^{k-1} u^{i-1}\beta_{3i}\rangle \lneqq I$.

Here, $\beta_{lm} = g_{lm} g_{12}^{-1}$ where $m = \{3, \ldots, k\}$ for $l = 1$, $m = \{1, \ldots, k\}$ for $l = 2$, and $m = \{1, \ldots, k-1\}$ for $l = 3$. Now, similar to cases 1 and 2, we have

$$gv = v \sum_{i=2}^{k} u^{i-1} g_{1i} + v^2 \sum_{i=1}^{k} u^{i-1} g_{2i}.$$

Hence,

$$gv - v^2 u^{k-1} g_{2k} = v \sum_{i=2}^{k} u^{i-1} g_{1i} + v^2 \sum_{i=1}^{k-1} u^{i-1} g_{2i} \ \in \ I$$

and further, we have following cases:

3. $\langle uv + v \sum_{i=3}^{k} u^{i-1} \beta_{1i} + v^2 \sum_{i=1}^{k-1} u^{i-1} \beta_{2i} \rangle = I$.
4. $\langle uv + v \sum_{i=3}^{k} u^{i-1} \beta_{1i} + v^2 \sum_{i=1}^{k-1} u^{i-1} \beta_{2i} \rangle \lneqq I$.

Here, $\beta_{lm} = g_{lm} g_{12}^{-1}$ where $m = \{3, \ldots, k\}$ for $l = 1$, $m = \{1, \ldots, k-1\}$ for $l = 2$. Now, similar to cases 1-4 above, we have

$$gv^2 = v^2 \sum_{i=2}^{k} u^{i-1} g_{1i}.$$

Hence,

$$gv^2 - v^2 u^{k-1} f_{1k} = v^2 \sum_{i=2}^{k-1} u^{i-1} g_{1i} \ \in \ I$$

and we have

5. $\langle uv^2 + v^2 \sum_{i=3}^{k-1} u^{i-1} \beta_{1i} \rangle = I$.

where $\beta_{lm} = g_{lm} g_{12}^{-1}$ for $m = \{3, \ldots, k-1\}$ and $l = 1$. □

Now, similar to Theorem 4, we have

Theorem 7 *Let $\mathcal{P}$ be a family of irreducible pairwise coprime monic polynomials P_i ($0 \le i \le 4k+2$) in $\mathbb{F}_{p^3}[x]$ such that $\prod_{i=0}^{4k+2} P_i = x^n - 1$, $\beta_{11} = 0$, and $\beta_{21} = 1$. Then a cyclic code $\mathcal{C}$ over $\mathcal{S}$ can be written as*

$$\mathcal{C} = \bigoplus_{i=1}^{k} \langle u^{i-1} \hat{P}_i \rangle \bigoplus_{i=1}^{k} \langle u^{i-1} v \hat{P}_{k+i} \rangle \bigoplus_{i=1}^{k} \langle u^{i-1} v^2 \hat{P}_{2k+i} \rangle \bigoplus_{i=1}^{k-1} \langle (u^i + v\eta_{1i} + v^2 \eta_{2i}) \hat{P}_{3k+i} \rangle$$

$$\bigoplus \langle (u + \sum_{i=3}^{k} u^{i-1} \beta_{1i} + v \sum_{i=1}^{k} u^{i-1} \beta_{2i} + v^2 \sum_{i=1}^{k-1} u^{i-1} \beta_{3i}) \hat{P}_{4k} \rangle$$

$$\bigoplus \langle (uv + v \sum_{i=3}^{k} u^{i-1} \beta_{1i} + v^2 \sum_{i=1}^{k-1} u^{i-1} \beta_{2i}) \hat{P}_{4k+1} \rangle \bigoplus \langle (uv^2 + v^2 \sum_{i=3}^{k-1} u^{i-1} \beta_{1i}) \hat{P}_{4k+2} \rangle.$$

Moreover, $|\mathcal{C}| = p^{3\gamma}$ *where*

$$\begin{aligned}\gamma &= \left(\sum_{i=1}^{k}(3k-3(i-1))\deg(P_i)+(2k-2(i-1))\deg(P_{k+i})+(k-(i-1))\deg(P_{2k+i})\right)\\ &+\left(\sum_{i=1}^{k-1}(3k-i)\deg(P_{3k+i})\right)+(3k-1)\deg(P_{4k})\\ &+(2k-1)\deg(P_{4k+1})+(k-1)\deg(P_{4k+2}).\end{aligned}$$

Proof Follows similar to Theorem 4. □

From now onward, we will use the structure of $\mathcal{C}$ defined in Theorem 7 above whenever we suppose that $\mathcal{C}$ is a cyclic code of length n over $\mathcal{S}$.

4.2 Euclidean dual of cyclic codes over $\mathcal{S}$

Euclidean inner product of $a = (a_0, a_1, \ldots, a_{n-1})$, $b = (b_0, b_1, \ldots, b_{n-1}) \in \mathcal{S}^n$ is defined by $\langle a, b\rangle = a_0b_0 + a_1b_1 + \cdots + a_{n-1}b_{n-1}$. If $\langle a, b\rangle = 0$, then a and b are considered orthogonal. Recall that the Euclidean dual of a linear code $\mathcal{C}$ is defined as $\mathcal{C}^{\perp} = \{b \in \mathcal{S}^n \mid \langle a, b\rangle = 0, \ \forall\ a \in \mathcal{C}\}$. For a polynomial $f(x) = f_0 + f_1x + f_2x^2 + \cdots + f_{n-1}x^{n-1}$ with $f_{n-1} \neq 0$, its reciprocal is defined by $f^*(x) = x^{n-1}f(1/x) = f_{n-1} + f_{n-2}x + \cdots + f_0x^{n-1}$. Note that $\deg f^*(x) \leq \deg f(x)$. If $f_0 \neq 0$, $\deg f^*(x) = \deg f(x)$.

Theorem 8 *Let* $\mathcal{C}$ *be the cyclic code of length* n *over* $\mathcal{S}$*. Then*

$$\mathcal{C}^{\perp} = \langle \hat{P}_0^*\rangle \bigoplus_{i=2}^{k}\langle u^{k-(i-1)}\hat{P}_i^*\rangle \bigoplus_{i=1}^{k}\langle v^2\hat{P}_{k+i}^*\rangle + \langle u^{k-(i-1)}\hat{P}_{k+i}^*\rangle \bigoplus_{i=1}^{k}\langle v\hat{P}_{2k+i}^*\rangle + \langle u^{k-(i-1)}\hat{P}_{2k+i}^*\rangle$$

$$\bigoplus_{i=1}^{k-1}\langle u^{k-i}v^2\hat{P}_{3k+i}^*\rangle \bigoplus\langle v^2u^{k-1}\hat{P}_{4k}^*\rangle \bigoplus\langle v^2\hat{P}_{4k+1}^*\rangle + \langle vu^{k-1}\hat{P}_{4k+1}^*\rangle \bigoplus\langle v\hat{P}_{4k+2}^*\rangle + \langle u^{k-1}\hat{P}_{4k+2}^*\rangle.$$

Thus, $|\mathcal{C}^{\perp}| = p^{3\delta}$ *where*

$$\delta = (3k)\deg P_0 + \sum_{i=2}^{k}3(i-1)\deg(P_i) + \sum_{i=1}^{k}(k+2(i-1))\deg(P_{k+i}) + \sum_{i=1}^{k}(2k+(i-1))\deg(P_{2k+i})$$

$$+\sum_{i=1}^{k-1} i\deg(P_{3k+i}) + \deg(P_{4k}) + (k+1)\deg(P_{4k+1}) + (2k+1)\deg(P_{4k+2}).$$

Proof Let $\mathcal{C}$ be the cyclic code of length n over $\mathcal{S}$. Then, from Theorem 3, we have $|\mathcal{C}| = p^{3\gamma}$ where

$$\gamma = \left(\sum_{i=1}^{k}(3k-3(i-1))\deg(P_i)+(2k-2(i-1))\deg(P_{k+i})+(k-(i-1))\deg(P_{2k+i})\right)$$

$$+\left(\sum_{i=1}^{k-1}(3k-i)\deg(P_{43+i})\right)+(3k-1)\deg(P_{4k})+(2k-1)\deg(P_{4k+1})+(k-1)\deg(P_{4k+2}).$$

Now, we know that $|\mathcal{C}||\mathcal{C}^{\perp}| = p^{3(3kn)}$ where $n = \sum_{i=1}^{4k+2}\deg(P_i)$. Hence, $|\mathcal{C}^{\perp}| = p^{3\delta}$ where

$$\delta = (3k)\deg P_0+\sum_{i=2}^{k}3(i-1)\deg(P_i)+\sum_{i=1}^{k}(k+2(i-1))\deg(P_{k+i})+\sum_{i=1}^{k}(2k+(i-1))\deg(P_{2k+i})$$

$$+\sum_{i=1}^{k-1}i\deg(P_{3k+i})+\deg(P_{4k})+(k+1)\deg(P_{4k+1})+(2k+1)\deg(P_{4k+2}).$$

Now, consider

$$\mathcal{C}^* = \langle\hat{P}_0^*\rangle\bigoplus_{i=2}^{k}\langle u^{k-(i-1)}\hat{P}_i^*\rangle\bigoplus_{i=1}^{k}\langle v^2\hat{P}_{k+i}^*\rangle+\langle u^{k-(i-1)}\hat{P}_{k+i}^*\rangle\bigoplus_{i=1}^{k}\langle v\hat{P}_{2k+i}^*\rangle+\langle u^{k-(i-1)}\hat{P}_{2k+i}^*\rangle$$

$$\bigoplus_{i=1}^{k-1}\langle u^{k-i}v^2\hat{P}_{3k+i}^*\rangle\bigoplus\langle v^2u^{k-1}\hat{P}_{4k}^*\rangle\bigoplus\langle v^2\hat{P}_{4k+1}^*\rangle+\langle vu^{k-1}\hat{P}_{4k+1}^*\rangle\bigoplus\langle v\hat{P}_{4k+2}^*\rangle+\langle u^{k-1}\hat{P}_{4k+2}^*\rangle.$$

We can see that

$$u^{i-1}u^{k-(i-1)}\hat{P}_i\hat{P}_i^* = 0 \text{ for } i = \{2,\ldots,k\}.$$

$$u^{i-1}v(\langle v^2\rangle+\langle u^{k-(i-1)}\rangle)\hat{P}_{k+i}\hat{P}_{k+i}^* = 0 \text{ for } i = \{1,\ldots,k\}.$$

$$u^{i-1}v^2(\langle v\rangle+\langle u^{k-(i-1)}\rangle)\hat{P}_{2k+i}\hat{P}_{2k+i}^* = 0 \text{ for } i = \{1,\ldots,k\}.$$

$$(u^i+v\eta_{1i}+v^2\eta_{2i})u^{k-i}v^2\hat{P}_{3k+i}\hat{P}_{3k+i}^* = 0 \text{ for } i = \{1,\ldots,k-1\}.$$

$$\langle(u+\sum_{i=3}^{k}u^{i-1}\beta_{1i}+v\sum_{i=1}^{k}u^{i-1}\beta_{2i}+v^2\sum_{i=1}^{k-1}u^{i-1}\beta_{3i})\rangle v^2u^{k-1}\hat{P}_{4k}\hat{P_{4k}}^* = 0.$$

$$\langle(uv+v\sum_{i=3}^{k}u^{i-1}\beta_{1i}+v^2\sum_{i=1}^{k-1}u^{i-1}\beta_{2i})\rangle(\langle v^2\rangle+\langle vu^{k-1}\rangle)\hat{P_{4k+1}}\hat{P}_{4k+1}^* = 0.$$

$$\langle(uv^2+v^2\sum_{i=3}^{k-1}u^{i-1}\beta_{1i})\rangle(\langle v\rangle+\langle u^{k-1}\rangle)\hat{P_{4k+2}}\hat{P}_{4k+2}^* = 0.$$

Hence, $\mathcal{C}^* \subseteq \mathcal{C}$.

Now,

$$|\hat{P_0}^*| = p^{3((3k)\deg P_0)}.\ |u^{k-(i-1)}\hat{P}_i^*| = p^{3(3(i-1)\deg(P_i))} \text{ for } i = \{2,\ldots,k\}.$$

$$|\hat{P}_{k+i}^*| = p^{3((k+2(i-1))\deg(P_{k+i}))}, |\hat{P}_{2k+i}^*| = p^{3((2k+(i-1))\deg(P_{2k+i}))} \text{ for } i = \{1,\ldots,k\}.$$

$$|\hat{P}_{3k+i}^*| = p^{3((i)\deg(P_{3k+i}))} \text{ for } i = \{1,\ldots,k-1\}.$$

$$|\hat{P}^*_{4k}| = p^{3(\deg(P_{4k}))},\ |\hat{P}^*_{4k+1}| = p^{3((k+1)\deg(P_{4k+1}))},\ |\hat{P}^*_{4k+2}| = p^{3((2k+1)\deg(P_{4k+2}))}.$$

Thus, $|c^*| = p^{3\delta}$ where

$$\begin{aligned}\delta &= (3k)\deg P_0 + \sum_{i=2}^{k} 3(i-1)\deg(P_i) + \sum_{i=1}^{k}(k+2(i-1))\deg(P_{k+i}) \\ &+ \sum_{i=1}^{k}(2k+(i-1))\deg(P_{2k+i}) + \sum_{i=1}^{k-1} i\deg(P_{3k+i}) \\ &+ \deg(P_{4k}) + (k+1)\deg(P_{4k+1}) + (2k+1)\deg(P_{4k+2}).\end{aligned}$$

Consequently, we get $|\mathcal{C}^*| = |\mathcal{C}^{\perp}|$.

Theorem 9 *Let $\mathcal{C}$ be a cyclic code of length n over $\mathcal{S}$. Then there exist polynomials $B^*_1, B^*_2, \ldots, B^*_{4k+2}$ in $\mathbb{F}_{27}$, which are factors of $x^n - 1$ such that*

$$\mathcal{C}^{\perp} = \bigoplus_{i=1}^{k} u^{i-1}\langle B^*_i\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{k} u^{i-1}v\langle B^*_{k+i}\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{k} u^{i-1}v^2\langle B^*_{2k+i}\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{k-1}(u^i + v\eta_{1i} + v^2\eta_{2i})\langle B^*_{3k+i}\rangle_{\mathcal{R}'}$$

$$\bigoplus(u + \sum_{i=3}^{k} u^{i-1}\beta_{1i} + v\sum_{i=1}^{k} u^{i-1}\beta_{2i} + v^2\sum_{i=1}^{k-1} u^{i-1}\beta_{3i})\langle B^*_{4k}\rangle_{\mathcal{R}'}$$

$$\bigoplus(uv + v\sum_{i=3}^{k} u^{i-1}\beta_{1i} + v^2\sum_{i=1}^{k-1} u^{i-1}\beta_{2i})\langle B^*_{4k+1}\rangle_{\mathcal{R}'} \bigoplus(uv^2 + v^2\sum_{i=3}^{k-1} u^{i-1}\beta_{1i})\langle B^*_{4k+2}\rangle_{\mathcal{R}'}$$

where β_{ij} and η_{mn} are units in $\mathbb{F}_{27}[x]$ and $\langle - \rangle$ is the ideal of $\mathcal{R}'$ generated by $-$. Also, we have $|\mathcal{C}^{\perp}| = p^{3((4k+2)n - (\deg(A^*_1) + \deg(A^*_2) + \cdots + \deg(A^*_{4k+2})))}$.

Proof It follows from Theorem 5 and Theorem 8. □

4.3 Hermitian dual of cyclic codes over $\mathcal{S}$

The Hermitian inner product of $a = (a_0, a_1, \ldots, a_{n-1})$ and $b = (b_0, b_1, \ldots, b_{n-1}) \in \mathcal{S}^n$ is defined by $\langle a, b\rangle_H = a_0\overline{b}_0 + a_1\overline{b}_1 + \cdots + a_{n-1}\overline{b}_{n-1}$ where "$-$" stands for conjugation and $\overline{b_i} = b_i^4$. For a code $\mathcal{C}$, its Hermitian dual $\mathcal{C}^{\perp}$ is defined as $\mathcal{C}^{\perp H} = \{b \in \mathcal{S}^n \mid \langle a, b\rangle_H = 0\ \forall\ a \in \mathcal{C}\}$. For a polynomial $f(x) = f_0 + f_1x + f_2x^2 + \cdots + f_{n-1}x^{n-1}$ with $f_{n-1} \neq 0$, its reciprocal is defined as $f^*(x) = x^{n-1}f(1/x) = f_{n-1} + f_{n-2}x + \cdots + f_0x^{n-1}$. It can be seen that $\deg f^*(x) \leq \deg f(x)$. If $f_0 \neq 0$, then $\deg f^*(x) = \deg f(x)$. We denote $\overline{f(x)} = f_0^4 + f_1^4x + f_2^4x^2 + \cdots + f_{n-1}^4x^{n-1}$.

Theorem 10 *Let $\mathcal{C}$ be the cyclic code of length n over $\mathcal{S}$. Then*

$$\mathcal{C}^{\perp_{\mathcal{H}}} = \langle\hat{\hat{P}}^*_0\rangle \bigoplus_{i=2}^{k}\langle u^{k-(i-1)}\hat{\hat{P}}^*_i\rangle \bigoplus_{i=1}^{k}\langle v^2\hat{\hat{P}}^*_{k+i}\rangle + \langle u^{k-(i-1)}\hat{\hat{P}}^*_{k+i}\rangle \bigoplus_{i=1}^{k}\langle v\hat{\hat{P}}^*_{2k+i}\rangle + \langle u^{k-(i-1)}\hat{\hat{P}}^*_{2k+i}\rangle$$

$$\bigoplus_{i=1}^{k-1}\langle u^{k-i}v^2\hat{\bar{P}}^*_{3k+i}\rangle \bigoplus\langle v^2u^{k-1}\hat{\bar{P}}^*_{4k}\rangle \bigoplus\langle v^2\hat{\bar{P}}^*_{4k+1}\rangle + \langle vu^{k-1}\hat{\bar{P}}^*_{4k+1}\rangle \bigoplus\langle v\hat{\bar{P}}^*_{4k+2}\rangle + \langle u^{k-1}\hat{\bar{P}}^*_{4k+2}\rangle.$$

Thus, $|\mathcal{C}^{\perp_{\mathcal{H}}}| = p^{3\Delta}$ *where*

$$\Delta = (3k)\deg P_0 + \sum_{i=2}^{k} 3(i-1)\deg(P_i) + \sum_{i=1}^{k}(k+2(i-1))\deg(P_{k+i}) + \sum_{i=1}^{k}(2k+(i-1))\deg(P_{2k+i})$$

$$+ \sum_{i=1}^{k-1} i\deg(P_{3k+i}) + \deg(P_{4k}) + (k+1)\deg(P_{4k+1}) + (2k+1)\deg(P_{4k+2}).$$

Proof It follows the same procedure as the proof of Theorem 8.

Theorem 11 *Let* $\mathcal{C}$ *be a cyclic code of length* n *over* $\mathcal{S}$*. Then there exist polynomials* $\bar{B}^*_1, \bar{B}^*_2, \ldots, \bar{B}^*_{4k+2}$ *in* $\mathbb{F}_{27}$*, which are factors of* $x^n - 1$ *such that*

$$\mathcal{C}^{\perp_{\mathcal{H}}} = \bigoplus_{i=1}^{k} u^{i-1}\langle \bar{B}^*_i\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{k} u^{i-1}v\langle \bar{B}^*_{k+i}\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{k} u^{i-1}v^2\langle \bar{B}^*_{2k+i}\rangle_{\mathcal{R}'} \bigoplus_{i=1}^{k-1}\left(u^i + v\eta_{1i} + v^2\eta_{2i}\right)\langle \bar{B}^*_{3k+i}\rangle_{\mathcal{R}'}$$

$$\bigoplus(u + \sum_{i=3}^{k} u^{i-1}\beta_{1i} + v\sum_{i=1}^{k} u^{i-1}\beta_{2i} + v^2\sum_{i=1}^{k-1} u^{i-1}\beta_{3i})\langle \bar{B}^*_{4k}\rangle_{\mathcal{R}'}$$

$$\bigoplus(uv + v\sum_{i=3}^{k} u^{i-1}\beta_{1i} + v^2\sum_{i=1}^{k-1} u^{i-1}\beta_{2i})\langle \bar{B}^*_{4k+1}\rangle_{\mathcal{R}'} \bigoplus(uv^2 + v^2\sum_{i=3}^{k-1} u^{i-1}\beta_{1i})\langle \bar{B}^*_{4k+2}\rangle_{\mathcal{R}'}$$

where β_{ij} *and* η_{mn} *are units in* $\mathbb{F}_{27}[x]$ *and* $\langle - \rangle$ *is the ideal of* $\mathcal{R}'$ *generated by* $-$*. Also we have,* $|\mathcal{C}^{\perp_{\mathcal{H}}}| = p^{3((4k+2)n - (\deg(A^*_1)+\deg(A^*_2)+\cdots+\deg(A^*_{4k+2})))}$.

Proof It follows from Theorem 5 and Theorem 10. □

5 Examples

5.1 Cyclic codes over $\mathcal{R} = M_3(\mathbb{F}_5 + u\mathbb{F}_5 + \cdots + u^{k-1}\mathbb{F}_5)$

Recall that a linear q-ary $[n, K, d]$ code is said to be MDS (maximum distance separable) if $d = n - K + 1$, and AMDS (almost maximum distance separable) if $d = n - K$. One of the fundamental problems for a linear code $[n, K]$ is to optimize d and obtain $d_{MDS} = n - K + 1$. In the following examples, we obtained linear codes over $\mathbb{F}_{125}$ for $p = 5$. Moreover, the codes we obtain over $\mathbb{F}_{125}$ are AMDS, and we use the Magma software [5] to carry out the computations for these examples.

Example 1 For $n = 2$, we have $x^2 - 1 = (x+1)(x+4)$ in $\mathbb{F}_{125}[x]$. Let $f_1 = x + 1$, $f_2 = x + 4$. For $p = 5$, $k = 2$, and $n = 2$, we obtain some cyclic codes of length 2 over $\mathcal{R}$ and their Gray images in the Table 1.

Example 2 For $n = 3$, we have $x^3 - 1 = (x+4)(x^2+x+1)$ in $\mathbb{F}_{125}[x]$. Let $f_1 = x+4$ and $f_2 = x^2 + x + 1$. We obtain cyclic codes of length 3 over $\mathcal{R}$ for $p = 5$, $k = 1, 2$, and $n = 3$, along with their Gray images as shown in the Table 1.

Example 3 For $n = 7$, we have $x^7 - 1 = (x+4)(x^2+w^{42}x+1)(x^2+w^{58}x+1)(x^2+w^{86}x+1)$ in $\mathbb{F}_{125}[x]$. Let $f_1 = x+4$, $f_2 = x^2+w^{42}x+1$, $f_3 = x^2+w^{58}x+1$, $f_4 = x^2 + w^{86}x + 1$. In the Table 1, we obtain cyclic codes of length 7 over $\mathcal{R}$ together with their Gray images for $p = 5$, $k = 1$, and $n = 7$.

5.2 Cyclic codes over $\mathcal{R} = M_3(\mathbb{F}_3 + u\mathbb{F}_3 + \cdots + u^{k-1}\mathbb{F}_3)$

There is a well-known online database, MinT [22], which provides bounds on the parameters of several linear codes over $\mathbb{F}_q$ where $q = p^n$. It is one of the most popular and reliable platforms to judge whether a linear code is better. Here, based on the optimal, best-known linear codes (briefly, BKLC) in the database [22], we construct many near optimal or near BKLC (NBKLC) codes over $\mathbb{F}_{27}$ such that $d - d' \leq 2$. We use the Magma software [5] to carry out the computation for these examples.

Example 4 For $n = 2$, we have $x^2 - 1 = (x+1)(x+2)$ in $\mathbb{F}_{27}[x]$. Let $f_1 = x+1$, $f_2 = x + 2$. For $p = 3$, $k = 2, 3$, and $n = 2$, we obtain some cyclic codes of length 2 over $\mathcal{R}$ and their Gray images in the Table 2.

Example 5 For $n = 5$, we have $x^5 - 1 = (x + 2)(x^4 + x^3 + x^2 + x + 1)$ in $\mathbb{F}_{27}[x]$. Let $f_1 = x + 2$, $f_2 = x^4 + x^3 + x^2 + x + 1$. In the Table 2, for $p = 3$, $k = 1$, and $n = 5$, we obtain some cyclic codes of length 5 over $\mathcal{R}$ along with their Gray images.

Example 6 For $n = 7$, we have $x^7 - 1 = (x+2)(x^2+w^5x+1)(x^2+w^{15}x+1)(x^2+w^{19}x+1)$ in $\mathbb{F}_{27}[x]$. Let $f_1 = x+2$, $f_2 = x^2+w^5x+1$, $f_3 = x^2+w^{15}x+1$, $f_4 = x^2 + w^{19}x + 1$. We obtain some cyclic codes of length 7 over $\mathcal{R}$ for $p = 3$, $k = 1$, and $n = 7$ along with their Gray images in the Table 2.

Example 7 For $n = 4$, we have $x^4 - 1 = (x + 1)(x + 2)(x^2 + 1)$ in $\mathbb{F}_{27}[x]$. Let $f_1 = x + 1$, $f_2 = x + 2$, $f_3 = x^2 + 1$. For $p = 3$, $k = 2, 3$, and $n = 4$, we obtain some cyclic codes of length 4 over $\mathcal{R}$ and their Gray images in the Table 2.

5.3 Cyclic codes over $\mathcal{R} = M_3(\mathbb{F}_2 + u\mathbb{F}_2 + \cdots + u^{k-1}\mathbb{F}_2)$

There is a well-known online database [11] that provides bounds on the parameters of several linear codes over $\mathbb{F}_q$, where $q = p^n \leq 9$. Here, based on the optimal, best-known (briefly, BKLC) codes in the database [11], we construct many near optimal or near BKLC codes over $\mathbb{F}_8$ such that $d - d' \leq 2$. We use the Magma software [5] to carry out the computation for these examples.

Example 8 For $n = 3$, we have $x^3 - 1 = (x + 1)(x^2 + x + 1)$. Let $g_1 = x + 1$, $g_2 = x^2 + x + 1$ in $\mathbb{F}_8[x]$. For $n = 7$, we have $x^7 - 1 = (x + 1)(x + w)(x + w^2)(x +$

Table 1 Gray images of cyclic codes over $\mathcal{R} = M_3(\mathbb{F}_5 + u\mathbb{F}_5 + \cdots + u^{k-1}\mathbb{F}_5)$

k	n	Generators of cyclic codes	Images under θ	Remark
1	3	$\langle (w^2e^2 + we + 2w^{10})f_1 \rangle$	$[9, 4, 5]_{125}$	AMDS
1	3	$\langle (w^2e^2 + we + 2w^{10})f_1, (w^5e^2 + w^7e + w)f_2 \rangle$	$[9, 5, 4]_{125}$	AMDS
1	3	$\langle (w^2e^2 + we + 2w^{10})f_1, (w^5e^2 + w^7e + 4w^{76})f_2 \rangle$	$[9, 6, 3]_{125}$	AMDS
2	2	$\langle (w^3v + w^5u + w^2v^2)f_1, (w^6v + w^9u + w^{11}v^2)f_2 \rangle$	$[12, 6, 6]_{125}$	AMDS
2	2	$\langle (w^3v + wv^2)f_1, (w^{11}v + w^5v^2)f_2 \rangle$	$[12, 8, 4]_{125}$	AMDS
2	2	$\langle (w^3v + wv^2)f_1, (w^{11}v)f_2 \rangle$	$[12, 9, 3]_{125}$	AMDS
2	2	$\langle (wv + w^2v^2)f_1, (w^3u + w^5v^2)f_2 \rangle$	$[12, 7, 5]_{125}$	AMDS
2	3	$\langle (w^3ue^2 + w^5ue + w^{23}u + 4w^3e^2 + 4w^3e + 4w^{13})f_1,$ $(w^{23}ue^2 + 4ew^{23} + 4e^2w^{23} + uw^{31} + 4w^7)f_2 \rangle$	$[18, 14, 4]_{125}$	AMDS
2	3	$\langle (w^3ue^2 + w^5ue + w^2 3u + 4w^3e^2 + 4w^3e + 4w^1 3)f_1,$ $(w^2 3ue^2 + 4ew^2 3 + 4e^2w^2 3 + 4uw^2 3 + 4w^1 16)f_2 \rangle$	$[18, 15, 3]_{125}$	AMDS
1	7	$\langle (w^{13}e^2 + w^{17}e + w^{19})f_1f_3f_4,$ $(w^5e^2 + w^7e + w^{11})f_2 \rangle$	$[21, 17, 4]_{125}$	AMDS
1	7	$\langle (w^{13}e^2 + w^{17}e + 4w^{99})f_2f_3f_4,$ $(w^5e^2 + w^7e + w^{11})f_1 \rangle$	$[21, 18, 3]_{125}$	AMDS

Table 2 Gray images of cyclic codes over $\mathcal{R} = M_3(\mathbb{F}_3 + u\mathbb{F}_3 + \cdots + u^{k-1}\mathbb{F}_3)$

k	n	Generators of cyclic codes	Images under θ	Remark
2	2	$\langle w^5 e f_1, (u + we + w^2e^2) f_2 \rangle$	$[12, 8, 4]_{27}$	AMDS
2	2	$\langle w^5 e f_2, (u + we + 2(1 + w)e^2) f_1 \rangle$	$[12, 9, 3]_{27}$	AMDS
1	5	$\langle (w^2e^2 + we + 2w^{10}) f_1, (w^5e^2 + w^7e + w) f_2 \rangle$	$[15, 12, 3]_{27}$	AMDS
3	2	$\langle (w^7ue + wu + w^4e + w^2e^2) f_1, (w^3e + w^5e^2) f_2 \rangle$	$[18, 12, 6]_{27}$	AMDS
3	2	$\langle (wu + w^4e + w^2e^2) f_1, (w^3e + w^5e^2) f_2 \rangle$	$[18, 13, 5]_{27}$	AMDS
3	2	$\langle (w^4e) f_1, (w^3e + w^5e^2) f_2 \rangle$	$[18, 15, 3]_{27}$	AMDS
1	7	$\langle (w^{17}e^2 + w^{19}e + w^{21}) f_2 + (w^{11}e^2 + w^3e + w^5) f_1 \rangle$	$[21, 13, 7]_{27}$	NBKLC
1	7	$\langle (w^{17}e^2 + w^{19}e + w^{21}) f_2 + (w^{11}e^2 + w^3e + w^5) f_1,$ $(w^{17}e^2 + w^{16}e + w^{15}) f_2 f_3 f_4 \rangle$	$[21, 14, 6]_{27}$	NBKLC
1	7	$\langle (w^{17}e^2 + w^{19}e + w^{21}) f_2, (w^5e^2 + w^7e + w) f_3 f_4 f_1 \rangle$	$[21, 17, 4]_{27}$	AMDS
1	7	$(w^{17}e^2 + w^{19}e + w^{21}) f_2, (w^5e^2 + w^7e + 2) f_3 f_4 f_1 \rangle$	$[21, 18, 3]_{27}$	AMDS
2	4	$\langle (wu + w^2e^2) f_1, (w^7ue + w^5u + w^3e + w^4e^2) f_3 \rangle$	$[24, 21, 3]_{27}$	AMDS
3	4	$\langle (wue + w^2u + 2w^{10}) f_1 f_2, (w^3e^2 + w^5e) f_3 \rangle$	$[36, 33, 3]_{27}$	AMDS

Table 3 Gray images of cyclic codes over $\mathcal{R} = M_3(\mathbb{F}_2 + u\mathbb{F}_2 + \cdots + u^{k-1}\mathbb{F}_2)$

k	n	Generators of cyclic codes	Images under θ
3	3	$\langle(wu^2e^2 + w^5u^2e + w^2e^2 + w^3e + w^4)g_1,$ $(w^2u^2e^2 + wue^2 + w^4ue + w^5u + w^5e^2 + w^6e + w^6)g_2\rangle$	$[27, 12, 10]_8$
3	3	$\langle(wu^2e^2 + w^5u^2e + w^2e^2 + w^3e + w^4)g_1,$ $(w^2u^2e^2 + wue^2 + w^5ue + w^5u + w^2e^2 + w^2e + w^4)g_2$ $+(wu^2e^2 + w^5u^2e + w^2e^2 + w^3e + w^4)g_1\rangle$	$[27, 14, 8]_8$
2	7	$\langle(w^2ue^2 + w^7e^2 + w^5e + w)f_6f_4f_1,$ $(w^2ue^2 + w^3e^2 + w^5)f_7f_2f_5f_3f_4\rangle$	$[42, 28, 7]_8$
2	7	$(w^2ue^2 + w^7e^2 + w^5e)f_6f_4, (w^2ue^2 + w^3e^2)f_7f_2\rangle$	$[42, 40, 2]_8$
2	9	$\langle(w^2ue^2 + w^7e^2 + w^5e + w^5)h_2,$ $(wue^2 + w^4ue + w^5u + w^2e^2 + w^6e + w)h_5h_3h_4h_1$	$[54, 45, 5]_8$

$w^3)(x + w^4)(x + w^5)(x + w^6)$. Let $f_1 = x + 1$, $f_2 = x + w$, $f_3 = x + w^2$, $f_4 = x + w^3$, $f_5 = x + w^4$, $f_6 = x + w^5$, $f_7 = x + w^6$ in $\mathbb{F}_8[x]$. For $n = 9$, we have $x^9 - 1 = (x + 1)(x^2 + x + 1)(x^2 + wx + 1)(x^2 + w^2x + 1)(x^2 + w^4x + 1)$. Let $h_1 = x+1$, $h_2 = x^2+x+1$, $h_3 = x^2+wx+1$, $h_4 = x^2+w^2x+1$, $h_5 = x^2+w^4x+1$ in $\mathbb{F}_8[x]$. For $p = 2, k = 2, 3$, and $n = 3, 7, 9$, we derive some near-optimal or near-BKLC cyclic codes of length 3, 7, and 9 over $\mathcal{R}$ and their Gray images in the Table 3.

6 Quantum codes

Recall that the set of n-fold tensor product $(\mathbb{C}^q)^{\otimes n} = \underbrace{\mathbb{C}^q \otimes \mathbb{C}^q \otimes \cdots \otimes \mathbb{C}^q}_{n \text{ times}}$ is a q^n-dimensional Hilbert space where $\mathbb{C}^q$ is a q-dimensional complex Hilbert space. Also, a q-ary quantum code of length n, dimension K and minimum distance d, denoted by $[[n, K, d]]_q$, is a q^K-dimensional subspace of $(\mathbb{C}^q)^{\otimes n}$ for which, the quantum Singleton defect [3] that measures how far a quantum code is from attaining the quantum Singleton bound ($2d \leq n - K + 2$) is defined as $\delta = n - K - 2d + 2$. Constructions in this paper yield $n - K$ to be even, making the defect δ also even. Hence, the smallest possible nonzero defects arising from our constructions are 0, 2 and 4. Quantum codes with $\delta = 0$ are quantum MDS codes. Motivated by this, we denote quantum MDS codes with $\delta = 0$ by $[[n, K, d]]_q^*$, quantum codes with $\delta = 2$ by $[[n, K, d]]_q^{**}$, and quantum codes with $\delta = 4$ by $[[n, K, d]]_q^{***}$. Also, a quantum code $[[n, K, d]]_q$ is said to be better than $[[n', K', d']]_q$ if any one of the following or both hold:

1. $d > d'$ when the code rate $\frac{K}{n} = \frac{K'}{n'}$ (Larger distance with same code rate).
2. $\frac{K}{n} > \frac{K'}{n'}$ when the distance $d = d'$ (Larger code rate with same distance).

6.1 CSS code framework

Calderbank, Shor, and Steane [7, 32] proposed a framework for constructing quantum error correction codes over qubits from two classical binary codes C_1 and C_2 satisfying $C_1^{\perp} \subseteq C_2$. This class of codes are called the *Calderbank-Shor-Steane (CSS) codes*. The condition $C_1^{\perp} \subseteq C_2$ is called the *dual-containing condition* of CSS codes. By considering the two codes C_1 and C_2 to be the same code, i.e., $C_1 = C_2$, we can construct quantum codes from dual-containing classical codes as $C_1^{\perp} \subseteq C_2 = C_1$.

The CSS codes form a class of stabilizer codes. Let H_1 and H_2 be the parity check matrices of the classical codes $C_1[n, K_1, d_1]$ and $C_2[n, K_2, d_2]$, respectively. As $C_1^{\perp} \subseteq C_2$, the elements of $C_1^{\perp}$ are codewords of C_2; hence, $H_2 H_1^{\mathrm{T}} = 0$.

The CSS code is defined in the following two equivalent ways:

1) The coset-based definition: As $C_1^{\perp} \subseteq C_2$, cosets of $C_1^{\perp}$ are formed in C_2. The basis codewords of the CSS code $\mathcal{Q}_{\mathrm{CSS}}$ are the normalized superposition of all the elements in a particular coset of $C_1^{\perp}$ in C_2. As $C_1^{\perp}$ has $2^{(n-K_1)}$ elements and C_2 has 2^{K_2} elements, we obtain C_2 to contain $(2^{K_2})/(2^{(n-K_1)}) = 2^{(K_1+K_2-n)}$ cosets of $C_1^{\perp}$. Also, each coset corresponds to a basis codeword, $\mathcal{Q}_{\mathrm{CSS}}$ with dimension $2^{(K_1+K_2-n)}$.

   Let $\omega_0, \omega_1, \ldots, \omega_{g-1}$ be the cosets of $C_1^{\perp}$ in C_2 where $g = 2^{(K_1+K_2-n)}$. Let $w_0, \ldots, w_{g-1}$ be the coset representatives of the cosets $\omega_0, \ldots, \omega_{g-1}$. The basis codeword of the CSS code corresponding to the coset ω_i ($i \in \{0, 1, \ldots, g-1\}$) is

$$|\psi_i\rangle = \frac{1}{2^{((n-K_1)/2)}} \sum_{l \in \mathcal{C}_1^{\perp}} |l + w_i\rangle. \tag{3}$$

   The basis states in the superposition help to detect/correct the bit flip errors, while the superposition helps to detect/correct the phase flip errors.
2) The check matrix based definition: The check matrix of the CSS code [23] is

$$\mathcal{H}_{\mathrm{CSS}} = \left[\begin{array}{c|c} H_1 & \mathbf{0} \\ \mathbf{0} & H_2 \end{array}\right]. \tag{4}$$

The quantum codes obtained from both these definitions are the same for qubits.

Let $\rho_1 = (n - K_1)$ and $\rho_2 = (n - K_2)$. From equation (4), the first ρ_1 stabilizer generators that correspond to $[H_1|0]$ operate only the bit flip operator on a few qubits. They do not operate phase flip operators. As the bit flip and phase flip operators do not commute with each other, these stabilizers are used to detect and correct the phase flip errors. Similarly, the stabilizers that correspond to $[0|H_2]$ detect and correct the bit flip errors.

As the stabilizer code [10] corrects bit flip errors and phase flip errors based on the stabilizers in $[0|H_2]$ and $[H_1|0]$, their bit flip and phase flip error correction capabilities are based on the error correction capabilities of H_2 and H_1, respectively. The minimum distance of the code is obtained to be $d' \geq \min(d_1, d_2)$ [23].

We assume that the parity check matrices H_1 and H_2 are full rank matrices. The check matrix in equation (4) is a $((\rho_1 + \rho_2) \times 2n)$ matrix. As H_1 and H_2 are full rank

matrices, the CSS code has $(\rho_1 + \rho_2)$ minimal stabilizer generators. Thus, the size of the CSS code is $2^{(n-(\rho_1+\rho_2))} = 2^{(K_1+K_2-n)}$. Hence, the CSS code is an $[[n, K_1 + K_2 - n, d' \geq \min(d_1, d_2)]]$ stabilizer code.

Remark 2 Using Euclidean or Hermitian inner product, we can study dual-containing codes over $\mathcal{S}$, but using the Gray map θ defined above, we cannot construct dual-containing codes over $\mathbb{F}_{27}$ which are necessary for constructing quantum codes via CSS construction. Hence, Similar to Euclidean dot product, we define $\langle a, b\rangle_R = \sum_{i=0}^{n-1} a_i b_{n-1-i}$ for $a, b \in \mathcal{S}^n$ which is a non-degenerate inner product. If $\langle a \cdot b\rangle_R = 0$, then a and b are considered orthogonal with respect to this inner product. We define the dual of a linear code $\mathcal{C}$ over $\langle\cdot\rangle_R$ as $\mathcal{C}^{\mathcal{R}\perp} = \{b \in \mathcal{S}^n \mid \langle a, b\rangle_R = 0,\ \forall\, a \in \mathcal{C}\}$.

Lemma 1 *$\forall x, y \in \mathcal{S}^n$, $\langle x, y\rangle_R = \sum_{i=0}^{n-1} x_i\, y_{n-1-i}$ is a non-degenerate inner product.*

Proof Let $x = (x_0, x_1, \ldots, x_{n-1}) \in \mathcal{S}^n$ be a non-zero vector. Then there exists an index i such that $x_i \neq 0$. As $\mathcal{S}$ contains 1, define $y = (y_0, y_1, \ldots, y_{n-1}) \in \mathcal{S}^n$ by setting $y_{n-1-i} = 1$ and $y_j = 0$ for all $j \neq n-1-i$. Then $\langle x, y\rangle_R = \sum_{k=0}^{n-1} x_k\, y_{n-1-k} = x_i \cdot 1 = x_i \neq 0$. Hence, for every non-zero $x \in \mathcal{S}^n$, there exists $y \in \mathcal{S}^n$ such that $\langle x, y\rangle_R \neq 0$. Thus, $\langle x, y\rangle_R$ is a non-degenerate inner product.

We introduce a new inner product to construct quantum codes with the help of matrix rings. This approach is novel in the literature, and we subsequently provided some better quantum codes that validate our approach.

Theorem 12 *Let $\mathcal{C}$ be the cyclic code of length n over $\mathcal{S}$. Then*

$$\mathcal{C}^{\mathcal{R}\perp} = \langle \hat{I}_0^* \rangle \bigoplus_{i=2}^{k} \langle u^{k-(i-1)} \hat{I}_i^* \rangle \bigoplus_{i=1}^{k} \langle v^2 \hat{I}_{k+i}^* \rangle + \langle u^{k-(i-1)} \hat{I}_{k+i}^* \rangle \bigoplus_{i=1}^{k} \langle v \hat{I}_{2k+i}^* \rangle + \langle u^{k-(i-1)} \hat{I}_{2k+i}^* \rangle$$

$$\bigoplus_{i=1}^{k-1} \langle u^{k-i} v^2 \hat{I}_{3k+i}^* \rangle \bigoplus \langle v^2 u^{k-1} \hat{I}_{4k}^* \rangle \bigoplus \langle v^2 \hat{I}_{4k+1}^* \rangle + \langle v u^{k-1} \hat{I}_{4k+1}^* \rangle \bigoplus \langle v \hat{I}_{4k+2}^* \rangle + \langle u^{k-1} \hat{I}_{4k+2}^* \rangle.$$

Thus, $|\mathcal{C}^{\perp}| = p^{3\delta}$ where

$$\delta = (3k) \deg I_0 + \sum_{i=2}^{k} 3(i-1) \deg(I_i) + \sum_{i=1}^{k} (k+2(i-1)) \deg(I_{k+i}) + \sum_{i=1}^{k} (2k+(i-1)) \deg(I_{2k+i})$$

$$+ \sum_{i=1}^{k-1} i \deg(I_{3k+i}) + \deg(I_{4k}) + (k+1) \deg(I_{4k+1}) + (2k+1) \deg(I_{4k+2}).$$

Proof Follows from Theorem 8 by reversing the index of $\hat{P}_i^*$, and then replacing $\hat{P}_i^*$ with $\hat{I}_{4k+2-i}^*$. □

Now, we provide the conditions under which $\mathcal{C}$ becomes the dual-containing code. We will consider a polynomial over $\mathbb{F}_{27}$ to be self-reciprocal if $f(x) = \alpha f^*(x)$ where α is a unit in $\mathbb{F}_{27}$.

Theorem 13 *If $x^n - 1 = f_1 f_2 \cdots f_m$ where f_i's are irreducible and self-reciprocal polynomials over $\mathbb{F}_{27}$, then a cyclic code of length n over $\mathcal{S} = M_3(\mathbb{F}_3 + u\mathbb{F}_3 + u^2\mathbb{F}_3 + \cdots + u^{k-1}\mathbb{F}_3)$ generated by ideals $\langle u\rangle f_i$, $\langle v\rangle f_j$, and $\langle (u + v\eta_1 + v^2\eta_2\rangle f_k$, where $\eta_1, \eta_2 \in \mathbb{F}_{27}$ is a dual-containing code.*

Proof From the Theorem 9, dual correspond to $\langle u\rangle f_i$ is $\langle u^{k-1}\rangle f_i^*$. Similarly, dual correspond to $\langle v\rangle f_j$, and $\langle (u+v\eta_1+v^2\eta_2\rangle f_k$ are $\langle v^2\rangle f_j^*$ and $\langle u^{k-1}v^2\rangle f_k^*$, respectively. From the given condition, $f_i = f_i^*$, $f_j = f_j^*$, and $f_k = f_k^*$. Also, it is easy to show that $\langle u^{k-1}\rangle \subseteq \langle u\rangle$, $\langle v^2\rangle \subseteq \langle v\rangle$, and $\langle u^{k-1}v^2\rangle \subseteq \langle u + v\eta_1 + v^2\eta_2\rangle$. Hence, over $\mathcal{S}$, we have $C^{\perp} \subseteq C$. Similarly, we can show that $C^{R\perp} \subseteq C$. □

6.2 Gray map for quantum codes

Construction of quantum codes from classical linear codes requires a dual-containing condition, i.e., $\mathcal{C}^{\perp} \subseteq \mathcal{C}$ over $\mathcal{S}$. In our case, we use $\mathcal{S} = M_3(\mathbb{F}_3 + u\mathbb{F}_3 + u^2\mathbb{F}_3 + \cdots + u^{k-1}\mathbb{F}_3)$, structure of $\mathcal{C}$ and $\mathcal{C}^{\mathcal{R}\perp}$ over $\mathcal{S}$ which we have studied above, and will show that $\phi(\mathcal{C}^{R\perp}) = \phi(\mathcal{C})^{R\perp}$ where $\phi = \phi_1$ is a Gray map from $\mathcal{S}$ to $\mathbb{F}_{27}^9$ for $k = 3$ and $\phi = \phi_2$ is a Gray map from $\mathcal{S}$ to $\mathbb{F}_{27}^6$ for $k = 2$ defined by

$$\phi_1[(d_1 + vd_2 + v^2d_3) + u(d_4 + vd_5 + v^2d_6) + u^2(d_7 + vd_8 + v^2d_9)]$$

$$= (d_9, d_8 + d_7, d_7, d_6, d_5, d_4, d_3 + d_2, d_2, d_1),$$

$$\phi_2[(d_1 + vd_2 + v^2d_3) + u(d_4 + vd_5 + v^2d_6)]$$

$$= (d_6, d_5 + d_4, d_4, d_3 + d_2, d_2, d_1),$$

where $d_i \in \mathbb{F}_{27}$ for all i. We then extend these Gray maps ϕ_1 and ϕ_2 componentwise to $\mathcal{S}^n$, where both ϕ_1 and ϕ_2 are distance preserving. The definition of Lee weight will follow a similar case to the case of the Gray map θ defined in Sect. 2. We choose different Gray map for quantum codes because θ does not follow the condition $\theta(\mathcal{C}^{R\perp}) = \theta(\mathcal{C})^{R\perp}$.

Theorem 14 *Let $\mathcal{C}$ be a cyclic code over $\mathcal{S}$ of length n, size M, and Lee distance d. Then $\phi(\mathcal{C})$ is a linear code over $\mathbb{F}_{27}$ of length $3kn$ for $k = 2, 3$, size M, and Hamming distance d.*

Now, we first show that above ϕ satisfies $\phi(\mathcal{C}^{R\perp}) = \phi(\mathcal{C})^{R\perp}$.

Theorem 15 *Let $\mathcal{C}$ be a cyclic code over $\mathcal{S}$ and $\mathcal{C}^{R\perp}$ be its dual. Then the Gray map ϕ satisfies $\phi(\mathcal{C}^{R\perp}) = \phi(\mathcal{C})^{R\perp}$.*

Proof First, we prove the theorem for $k = 2$. Let $a = (a_0, a_1, \ldots, a_n) \in \mathcal{C}$ where each $a_i = (d_1 + vd_2 + v^2d_3) + u(d_4 + vd_5 + v^2d_6)$ and $d_i \in \mathbb{F}_{27}$. Let $b = (b_0, b_1, \ldots, b_n) \in$

$\mathcal{C}^{\perp}$ where each $b_i = (c_1+vc_2+v^2c_3)+u(c_4+vc_5+v^2c_6)$ and $c_i \in \mathbb{F}_{27}$. By assuming $ab = 0$, we prove $\langle \phi(a)\phi(b)\rangle_R = 0$.

Now, from the Gray map defined above, $\langle \phi(a), \phi(b)\rangle_R = d_6c_1 + (d_5 + d_4)c_2 + d_4(c_3+c_2)+(d_3+d_2)c_4+d_2(c_5+c_4)+d_1c_6$. So, $\langle \phi(a), \phi(b)\rangle_R \neq 0$ when there exists some d_ic_j from the sum above such that $d_ic_j \neq 0$, which means that $r_id_iw_jc_j = 0$ but $d_ic_j \neq 0$ where r_i and w_j are corresponding component of the ring before applying a Gray map. This means that $r_iw_j = 0$ for some d_ic_j. But, $r_iw_j = uv^2$ for d_6c_1, $(uv + u)v$ for $(d_5 + d_4)c_2$, $u(v + v^2)$ for $d_4(c_3 + c_2)$, $(v + v^2)u$ for $(d_3 + d_2)c_4$, $v(uv + u)$ for $d_2(c_5 + c_4)$, and uv^2 for d_1c_6. Hence, $r_iw_j \neq 0$ for all d_ic_j in the summand. Thus, $\langle \phi(a), \phi(b)\rangle_R = 0$. Similarly, we prove the theorem for $k = 3$. □

Under the Gray maps ϕ_1 and ϕ_2 defined above, cyclic codes over $\mathcal{S}$ become linear codes over $\mathbb{F}_{27}$. Consequently, the newly defined inner product takes the form

$$\langle a, b\rangle_R = \sum_{i=0}^{n-1} a_i\, b_{n-1-i},\ \forall a, b \in \mathbb{F}_{27}^n.$$

Let R denote the coordinate reversal map, which maps b_i to b_{n-1-i}. Then $\langle a, b\rangle_R = \langle a, R(b)\rangle$, where $\langle a, R(b)\rangle$ is the standard Euclidean inner product. Towards quantum codes, this permutation corresponds to a relabeling of qubits ($i \mapsto n - 1 - i$), which does not change the error-correcting properties of the code or the underlying quantum channel. In the stabilizer formalism (as we discussed in the Sect. 6.1(2)), commutation between Z-type operators (Z(a)) and X-type operators (X(b)) is determined by the symplectic inner product [23, Secs. 10.4.2 and 10.5], which in this setting reduces to

$$(R(b)|0) \odot (0|a) = \langle R(b), a\rangle = \langle a, R(b)\rangle.$$

Thus, the condition ($\langle a, b\rangle_R = 0$) ensures that all corresponding stabilizer generators commute. Hence, the construction is fully compatible with the CSS framework, and is equivalent to a standard CSS code up to a relabeling of qubits.

Lemma 2 *[7] An $[[n, K, d]]_q$ quantum code satisfies the bound $2d \leq n - K + 2$, and is said to MDS (maximum-distance-separable) when $n + 2 = K + 2d$. This bound is known as the quantum Singleton Bound (QSB).*

Presently, the CSS construction (Lemma 3) is one of the widely used techniques for obtaining quantum codes from classical linear codes, in which dual-containing linear codes play an instrumental role.

Lemma 3 *[12, Theorem 3] Let $\mathcal{C}_1 = [n, K_1, d_1]_q$ and $\mathcal{C}_2 = [n, K_2, d_2]_q$ be two linear codes over $GF(q)$ with $\mathcal{C}_2^{\perp} \subseteq \mathcal{C}_1$. Then there exists a quantum error-correcting code $\mathcal{C} = [[n, K_1 + K_2 - n, d]]$ where $d = \min\{w(v) : v \in (\mathcal{C}_1\backslash\mathcal{C}_2^{\perp}) \cup (\mathcal{C}_2\backslash\mathcal{C}_1^{\perp})\} \geq \min\{d_1, d_2\}$. Further, if $\mathcal{C}_1^{\perp} \subseteq \mathcal{C}_1$, then there exists a quantum error-correcting code $\mathcal{C} = [[n, 2K_1 - n, d_1]]$ where $d_1 = \min\{w(v) : v \in \mathcal{C}_1\backslash\mathcal{C}_1^{\perp}\}$.*

Theorem 16 *Let $x^n - 1 = f_1 f_2 \cdots f_m$ where f_i's are irreducible and self-reciprocal polynomials over $\mathbb{F}_{27}$. Let $\mathcal{C}$ be a cyclic code of length of n over $\mathcal{S}$ generated by ideals*

Table 4 Obtained quantum codes via the CSS construction

k	n	Generators of cyclic codes	Images under θ	Quantum codes
2	2	$\langle w^5 v f_2, (u + w^7 v + w v^2) f_1 \rangle$	$[12, 8, 4]_{27}$	$[[12, 4, 4]]^{**}_{27}$
2	2	$\langle u f_2, w^2 v f_1 \rangle$	$[12, 10, 2]_{27}$	$[[12, 8, 2]]^{**}_{27}$
3	2	$\langle (u + 2v + w^5 v^2) f_1, w^7 v f_2 \rangle$	$[18, 15, 3]_{27}$	$[[18, 12, 3]]^{**}_{27}$
3	2	$\langle (u + w^3 v + w^5 v^2) f_2, (u + w^7 v + w^4 v^2) f_1 \rangle$	$[18, 14, 4]_{27}$	$[[18, 10, 4]]^{**}_{27}$
2	4	$\langle w^2 u f_3 f_2, (u + w^{17} v + w v^2) f_3 f_1,$ $(u + w^7 v + w^{11} v^2) f_2 f_1 \rangle$	$[24, 17, 6]_{27}$	$[[24, 10, 6]]^{***}_{27}$
2	4	$\langle w^2 u f_3 f_2, w^{17} v f_3 f_1,$ $(u + w^7 v + w^{11} v^2) f_2 f_1 \rangle$	$[24, 19, 4]_{27}$	$[[24, 14, 4]]^{***}_{27}$
2	4	$\langle w^2 v f_3, w^{17} v f_2 f_1,$ $(u + w^7 v + w^{11} v^2) f_3 f_1 \rangle$	$[24, 23, 2]_{27}$	$[[24, 22, 2]]^{*}_{27}$
3	4	$\langle w^2 v f_2 f_3, u w^6 f_3 f_1, w^3 u f_2 f_1 \rangle$	$[36, 35, 2]_{27}$	$[[36, 34, 2]]^{*}_{27}$
3	4	$\langle w^5 v f_2 f_3, (u + w^4 v + w^2 v^2) f_3 f_1,$ $(u + w^3 v + w^7 v^2) f_2 f_1 \rangle$	$[36, 31, 4]_{27}$	$[[36, 26, 4]]^{***}_{27}$
3	4	$\langle w^2 v f_2 f_3, u w^6 f_3 f_1,$ $(u + w^3 v + w^7 v^2) f_2 f_1 \rangle$	$[36, 33, 3]_{27}$	$[[36, 30, 3]]^{**}_{27}$

$\langle u\rangle f_i$, $\langle v\rangle f_j$, and $\langle (u+v\eta_1+v^2\eta_2)\rangle f_k$, where $\eta_1, \eta_2 \in \mathbb{F}_{27}$. Then there exists a quantum code with parameters $[[3kn, 2K-3kn, d_H]]_{27}$ for $k = 2, 3$.

Proof By Theorem 13, we have $\mathcal{C}^{R\perp} \subseteq \mathcal{C}$. Also, by Theorem 15, we have $\varphi(\mathcal{C}^{R\perp}) = \phi(\mathcal{C})^{R\perp}$. Therefore, $\varphi(\mathcal{C})^{R\perp} \subseteq \varphi(\mathcal{C})$. Hence, $\varphi(\mathcal{C})$ is a dual-containing linear code with parameters $[3kn, K, d_H]$ over $\mathbb{F}_{27}$. By Lemma 3, we conclude that there exists a quantum code with parameters $[[3kn, 2K-3kn, d_H]]_{27}$. □

7 Numerical illustrations

In this section, we construct quantum codes with the help of matrix rings. We use the factorization of x^n-1 for $n = 2$ and 4 as given in the examples 4 and 7. Here, $f_1^* = f_1$ and $f_2^* = f_2$ for the factorization of x^2-1, and $f_1^* = f_1$, $f_2^* = f_2$, and $f_3^* = f_3$ for the factorization of x^4-1, where f_i^* represents the reciprocal of f_i defined in the Sect. 4.2. First, we work over $k = 2$, with conditions $u^2 = v^3 = 0$.

Example 9 Consider a code $\mathcal{C}$ generated by $\langle w^5vf_2, (u+w^7v+wv^2)f_1\rangle$. From Theorem 16, $\mathcal{C}$ is a **dual-containing code** with parameters $[12, 8, 4]_{27}$. Hence, quantum code has parameters $[[12, 4, 4]]_{27}$. For this code, quantum Singleton defect $\delta = n-K-2d+2 = 12-4-2(4)+2 = 2$. Therefore, we denote this code by $[[12, 4, 4]]_{27}^{**}$. Similarly, we obtain other examples for $k = 2$ in the Table 4.

Now, we work over $k = 3$ and with the conditions $u^3 = v^3 = 0$.

Example 10 Consider a code $\mathcal{C}$ generated by $\langle (u+2v+w^5v^2)f_1, w^7vf_2\rangle$. From Theorem 16, $\mathcal{C}$ is a **dual-containing code** with parameters $[18, 15, 3]_{27}$. Hence, quantum code has parameters $[[18, 12, 3]]_{27}$. For this code, quantum Singleton defect $\delta = n-K-2d+2 = 18-12-2(3)+2 = 2$. Therefore, we denote this code by $[[18, 12, 3]]_{27}^{**}$. Similarly, we obtain other examples for $k = 3$ in the Table 4.

8 Conclusion

In 2024, Islam et al. [17] discussed the structure of cyclic codes over $M_3(\mathbb{F}_p)$, in which they provided many good linear codes over $\mathbb{F}_8$ with length $3n$ where $p \nmid n$. Initially, we have extended their work and studied the structure of cyclic codes over the ring $\mathcal{R}$ along with the Gray map, which takes our cyclic code over $\mathcal{R}$ to $\mathbb{F}_{p^3}$, with length $3kn$, where $p \nmid n$. We have shown that our structure is suitable for the construction of linear codes over $\mathbb{F}_{125}, \mathbb{F}_{27}$, as well as $\mathbb{F}_8$. Further, along with Euclidean and Hermitian duals, we have studied the dual-containing codes over the ring $\mathcal{S}$. Using a suitable Gray map, we have provided a novel method to study quantum codes over matrix rings via the CSS construction. Finally, we have provided many better quantum codes to validate our results. Here, the construction of self-dual and quantum codes over all possible primes remains open and presents an interesting problem for further study.

Acknowledgements The first and third authors are thankful to the Indian Institute of Technology Patna (India) for providing research facilities.

Author Contributions SRJ and SP initiated the work under the guidance of OP. Draft prepared by SRJ and reviewed by all authors. Finally submitted by OP.

Data Availability The authors declare that [the/all other] data supporting the findings of this study are available within the article. Any clarification may be requested from corresponding author provided it is essential.

Declarations

Conflict of interest The authors declare no conflict of interest.

References

1. Alahmadi, A., Islam, H., Prakash, O., Solé, P., Alkenani, A., Muthana, N., Hijazi, R.: New quantum codes from constacyclic codes over a non-chain ring. Quantum Inf. Process. **20**(2), 60 (2021)
2. Alahmadi, A., Sboui, H., Solé, P., Yemen, O.: Cyclic codes over $M_2(\mathbb{F}_2)$. J. Franklin Inst. **350**(9), 2837–2847 (2013)
3. Barbero-Lucas, B., Hernando, F., Martín-Cruz, H., McGuire, G.: MDS, Hermitian almost MDS, and Gilbert-Varshamov quantum codes from generalized monomial-Cartesian codes. Quantum Inf. Process. **23**(3), 86 (2024)
4. Bonnecaze, A., Udaya, P.: Cyclic codes and self-dual codes over $\mathbb{F}_2+u\mathbb{F}_2$. IEEE Trans. Inform. Theory **45**(4), 1250–1255 (1999)
5. Bosma, W., Cannon, J.: Handbook of Magma Functions. Univ. of Sydney, Sydney (1995)
6. Calderbank, A.R., Rains, E.M., Shor, P.W., Sloane, N.J.A.: Quantum error correction via codes over GF(4). IEEE Trans. Inf. Theory **44**(4), 1369–1387 (1998)
7. Calderbank, A.R., Shor, P.W.: Good quantum error-correcting codes exist. Phys. Rev. A **54**, 1098–1105 (1996)
8. Dertli, A., Cengellenmis, Y., Eren, S.: On quantum codes obtained from cyclic codes over A_2. Int. J. Quantum Inf. **13**(3), 1550031 (2015)
9. Falcunit, D. F., Sison, V. P.: Cyclic codes over matrix ring $M_2(\mathbb{F}_p)$ and their isometric images over $\mathbb{F}_{p^2}+u\mathbb{F}_{p^2}$. Int. Zurich Seminar Commun. (IZS), 26-28 (2014)
10. Gottesman, D.: Stabilizer codes and quantum error correction, Pasadena (1997). (Ph.D. thesis)
11. Grassl, M.: Bounds on the minimum distance of linear codes and quantum codes, (2007). (Online available at) http://www.codetables.de. Accessed 20 Aug 2025
12. Grassl, M., Beth, T., Roetteler, M.: On optimal quantum codes. Int. J. Quantum Inf. **2**(1), 55–64 (2004)
13. Greferath, M.: Cyclic codes over finite rings. Discrete Math. **177**(1–3), 273–277 (1997)
14. Hammons, A.R., Kumar, P.V., Calderbank, A.R., Sloane, N.J.A., Solé, P.: The $\mathbb{Z}_4$-linearity of Kerdock, Preparata, Goethals, and related codes. IEEE Trans. Inform. Theory **40**(2), 301–319 (1994)
15. Islam, H., Martínez-Moro, E., Prakash, O.: Cyclic codes over a non-chain ring $R_{e,q}$ and their application to LCD codes. Discrete Math. **344**(10), 112545 (2021)
16. Islam, H., Patel, S., Prakash, O.: A family of constacyclic codes over a class of non-chain rings $\mathcal{A}_{q,r}$ and new quantum codes. J. Appl. Math. Comput. **68**, 2493–2514 (2022)
17. Islam, H., Prakash, O., Bhunia, D.K.: Algebraic structure of cyclic codes over $M_3(\mathbb{F}_p)$. J. Discret. Math. Sci. Cryptogr. **27**, 189–201 (2024)
18. Islam, H., Prakash, O.: New quantum codes from constacyclic and additive constacyclic codes. Quantum Inf. Process. **19**(9), 319 (2020)
19. Joshi, S.R., Patel, S., Prakash, O.: Construction of new codes over $\mathbb{F}_{16}$ from cyclic codes over $\mathcal{M}_4(\mathbb{F}_2+u\mathbb{F}_2+u^2\mathbb{F}_2)$. In: Proc. IEEE Int. Symp. Inf. Theory (ISIT), Ann Arbor, MI, USA, 1–5 (2025)
20. Kai, X., Zhu, S.: Quaternary construction of quantum codes from cyclic codes over $\mathbb{F}_4+u\mathbb{F}_4$. Int. J. Quantum Inf. **9**(2), 689–700 (2011)
21. Luo, R., Udaya, P.: Cyclic codes over $M_2(\mathbb{F}_2+u\mathbb{F}_2)$. Cryptogr. Commun. **10**(6), 1109–1117 (2018)
22. MinT - A Database for Optimal Net, Sequence, Array, and Code Parameters. Austria (). (**MinT Project**) 7https://mint.sbg.ac.at/, last accessed 13 Aug 2025
23. Nielsen, M.A., Chuang, I.L.: Quantum Computation and Quantum Information, 10th edn. Cambridge University Press, Cambridge (2011)

24. Oggier, F., Solé, P., Belfiore, J.C.: Codes over matrix rings for space-time coded modulations. IEEE Trans. Inform. Theory **58**(2), 734–746 (2012)
25. Patel, S., Joshi, S.R., Prakash, O.: Construction of codes over $\mathbb{F}_{p^3}$ from cyclic codes over $M_3(\mathbb{F}_p + u\mathbb{F}_p + u^2\mathbb{F}_p)$. In: Proc. 2025 11th International Workshop on Signal Design and Its Applications in Communications (IWSDA), 1-5 (2025)
26. Patel, S., Prakash, O., Islam, H.: Cyclic codes over $M_4(\mathbb{F}_2 + u\mathbb{F}_2)$. Cryptogr. Commun. **14**(5), 1021–1034 (2022)
27. Pless, V., Qian, Z.: Cyclic codes and quadratic residue codes over $\mathbb{Z}_4$. IEEE Trans. Inform. Theory **42**(5), 1594–1600 (1996)
28. Prakash, O., Patel, S., Islam, H.: Quantum codes from skew cyclic codes over mixed alphabets. Comp. Appl. Math. **44**(8), 384 (2025)
29. Prakash, O., Verma, R.K., Singh, A.: Quantum and LCD codes from skew constacyclic codes over a finite non-chain ring. Quantum Inf. Process. **22**(5), 25 (2023)
30. Qian, J., Ma, W., Gou, W.: Quantum codes from cyclic codes over finite ring. Int. J. Quantum Inf. **7**(6), 1277–1283 (2009)
31. Shor, P.W.: Scheme for reducing decoherence in quantum computer memory. Phys. Rev. A **52**(4), R2493 (1995)
32. Steane, A.M.: Simple quantum error-correcting codes. Phys. Rev. A **54**, 4741–4751 (1996)
33. Wisbauer, R.: Foundations of Module and Ring Theory, Gordon and Breach (1991)
34. Zhu, S., Guo, H., Kai, X., Sun, Z.: New quantum codes derived from images of cyclic codes. Quantum Inf. Process. **21**(7), 254 (2022)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.